EDS LAB MATERIAL PORTAL

PHASE 1 WALKTHROUGH · ENGINEER TO LAB ADMIN

Gaurav Shelke · EDS Lab, Pune

October 2026

JCB

WHY WE BUILT IT

TODAY

- Stock checked by hand in the Excel sheet.
- Harness BOMs compared with stock line by line.
- Requests by mail or word of mouth; no status for the engineer.
- Nobody sees at a glance what is short and what is on order.

WITH THE PORTAL

- One place for live stock, calculated from every movement.
- Paste a BOM and see the gaps in seconds.
- Every request tracked from submission to release.
- Each role sees only its own pages: Engineer, Lab Admin, Lab Lead, Manager.

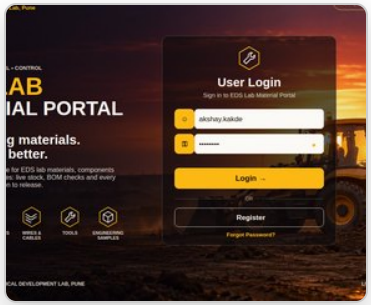
Runs in Power Apps on the lab's existing workbook. No new licence, no server; opens in the browser or Teams.

PHASE 1 AT A GLANCE

Engineer to Lab Admin in about 2 minutes, on live lab stock.

STEP 1

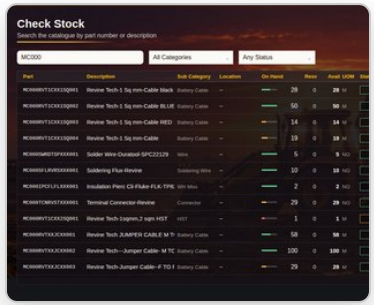
LOG IN



Username and password

STEP 2

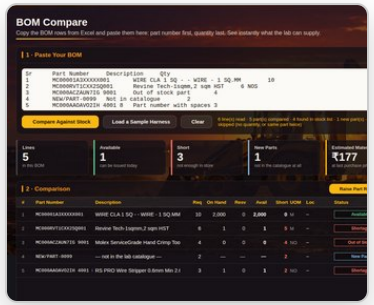
CHECK STOCK



Live on hand for every part

STEP 3

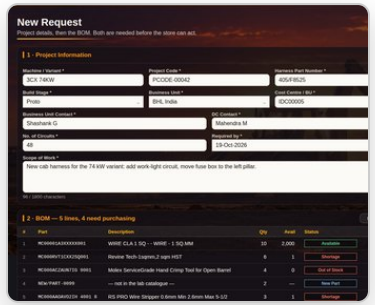
BOM COMPARE



Paste the Excel BOM, see the gaps

STEP 4

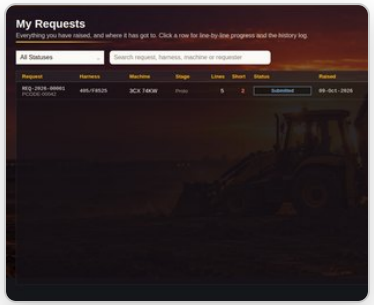
RAISE REQUEST



Project details and BOM lines

STEP 5

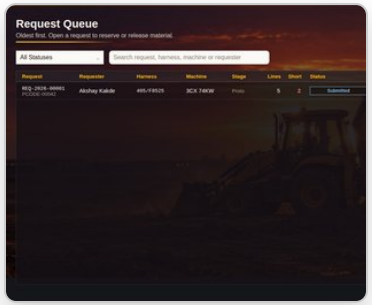
MY REQUESTS



Track status any time

STEP 6

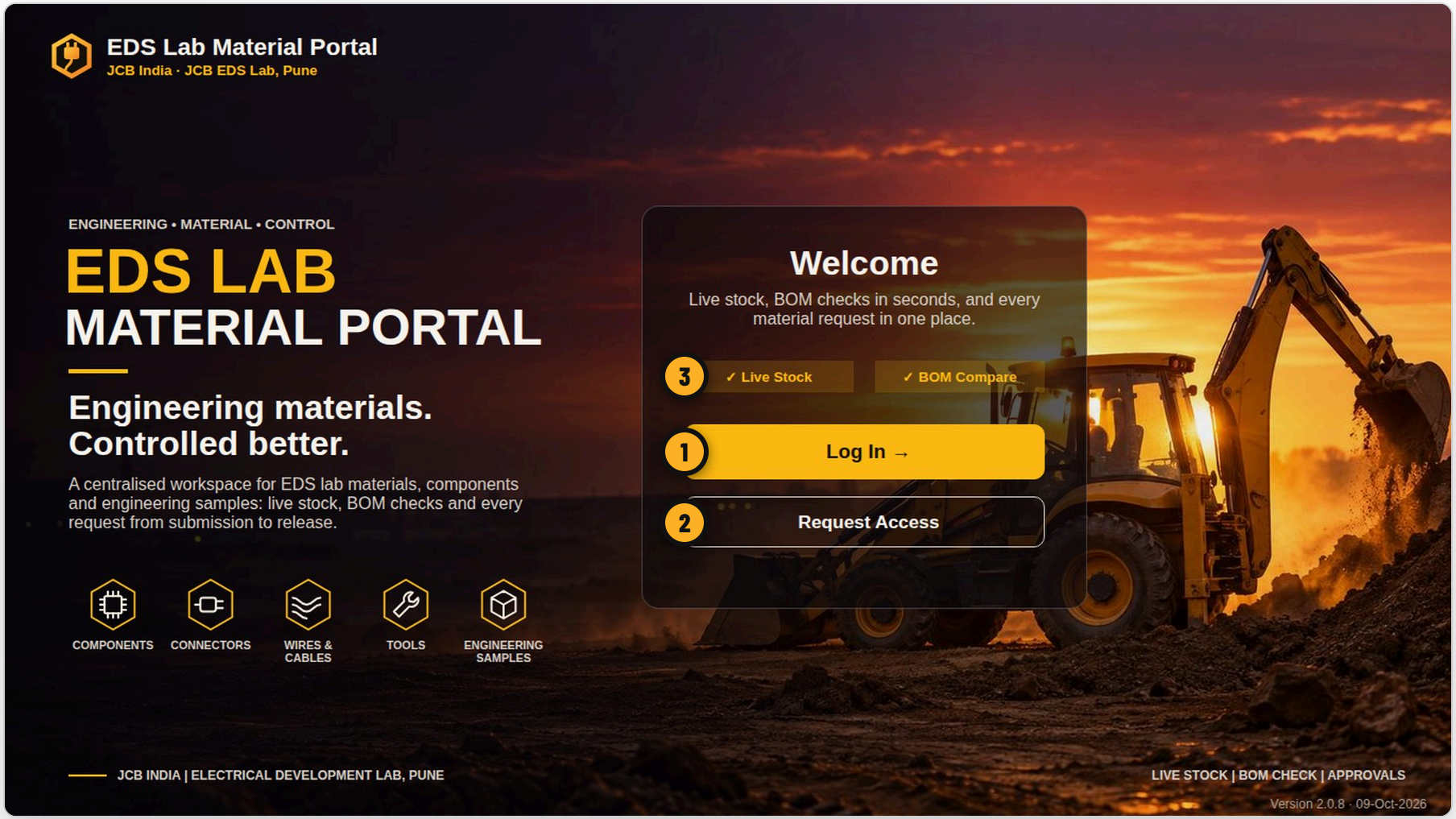
LAB ADMIN



Store sees it instantly

> Tested end to end on the lab's real data before every release: login, stock search, BOM paste, request saved, Lab Admin sees it.

WELCOME PAGE

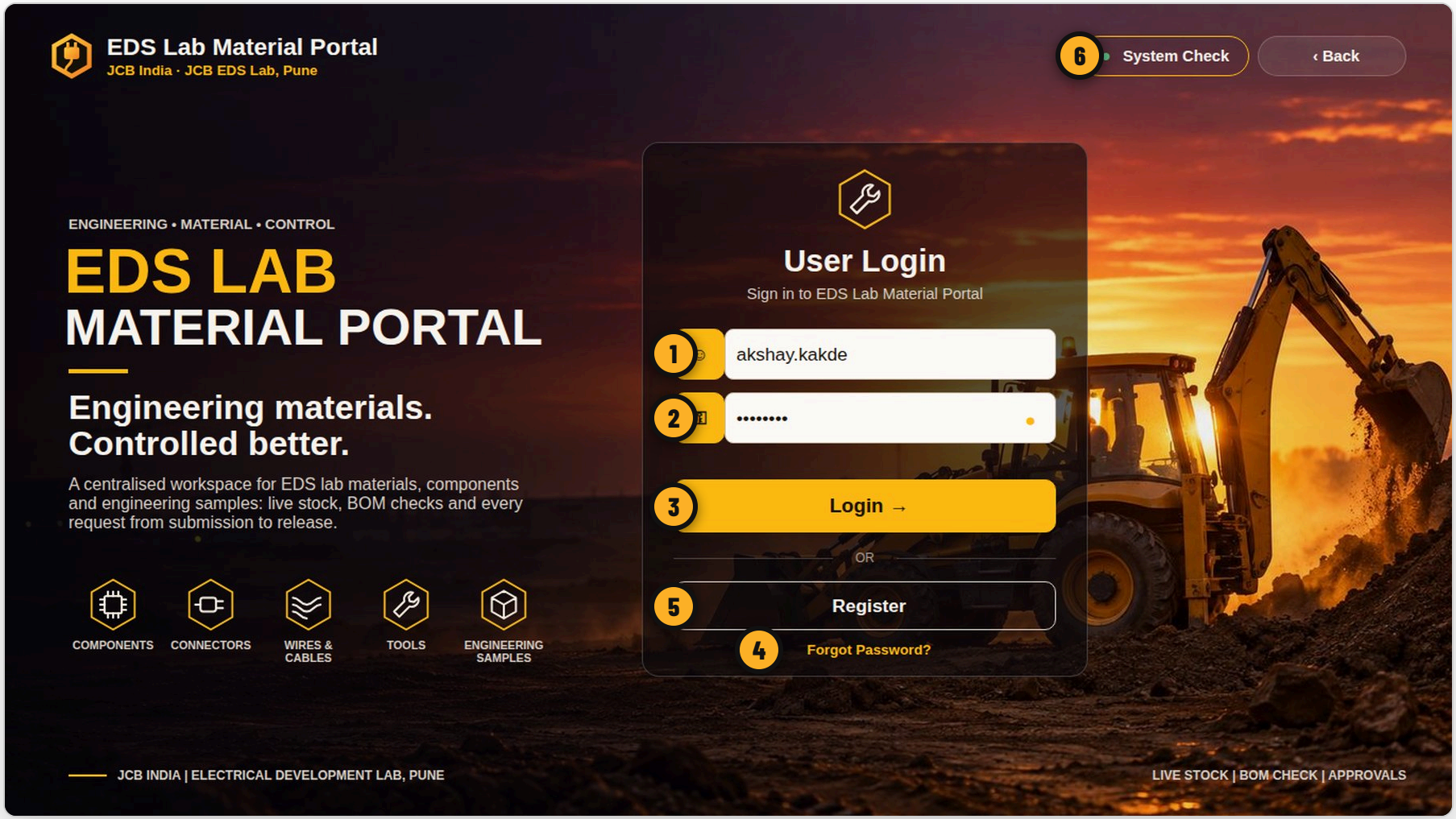


CLICK → WHAT HAPPENS

1. **Log In**
   Opens the login page.
2. **Request Access**
   A new person asks for an account; the Lab Admin gives the role.
3. **What the portal does**
   Live stock, BOM compare and request tracking, in one line.

> **RESULT**
> Nobody sees lab data before logging in.

LOGIN PAGE

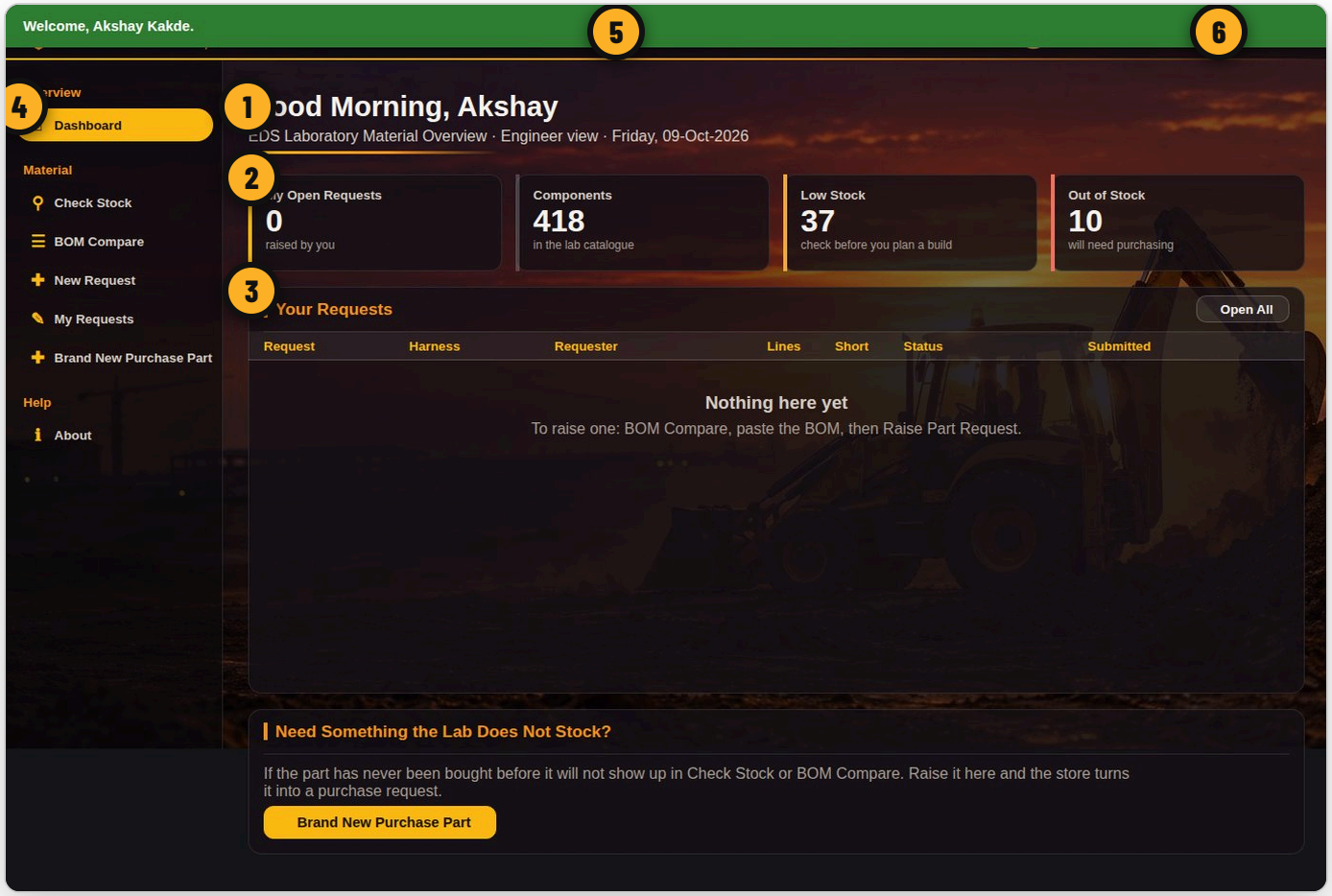


CLICK → WHAT HAPPENS

1. **Username**
   The name the Lab Admin set up, e.g. akshay.kakde.
2. **Password**
   Eye icon shows or hides it. Empty password? You choose one on first login.
3. **Login**
   Opens the dashboard for your role.
4. **Forgot Password?**
   Type your username and set a new password. No waiting for anyone.
5. **Register**
   New user fills name, username and password; role starts as Pending.
6. **System Check**
   18 health checks of the data links, for support.

RESULT
The engineer lands on the Engineer dashboard.

ENGINEER DASHBOARD

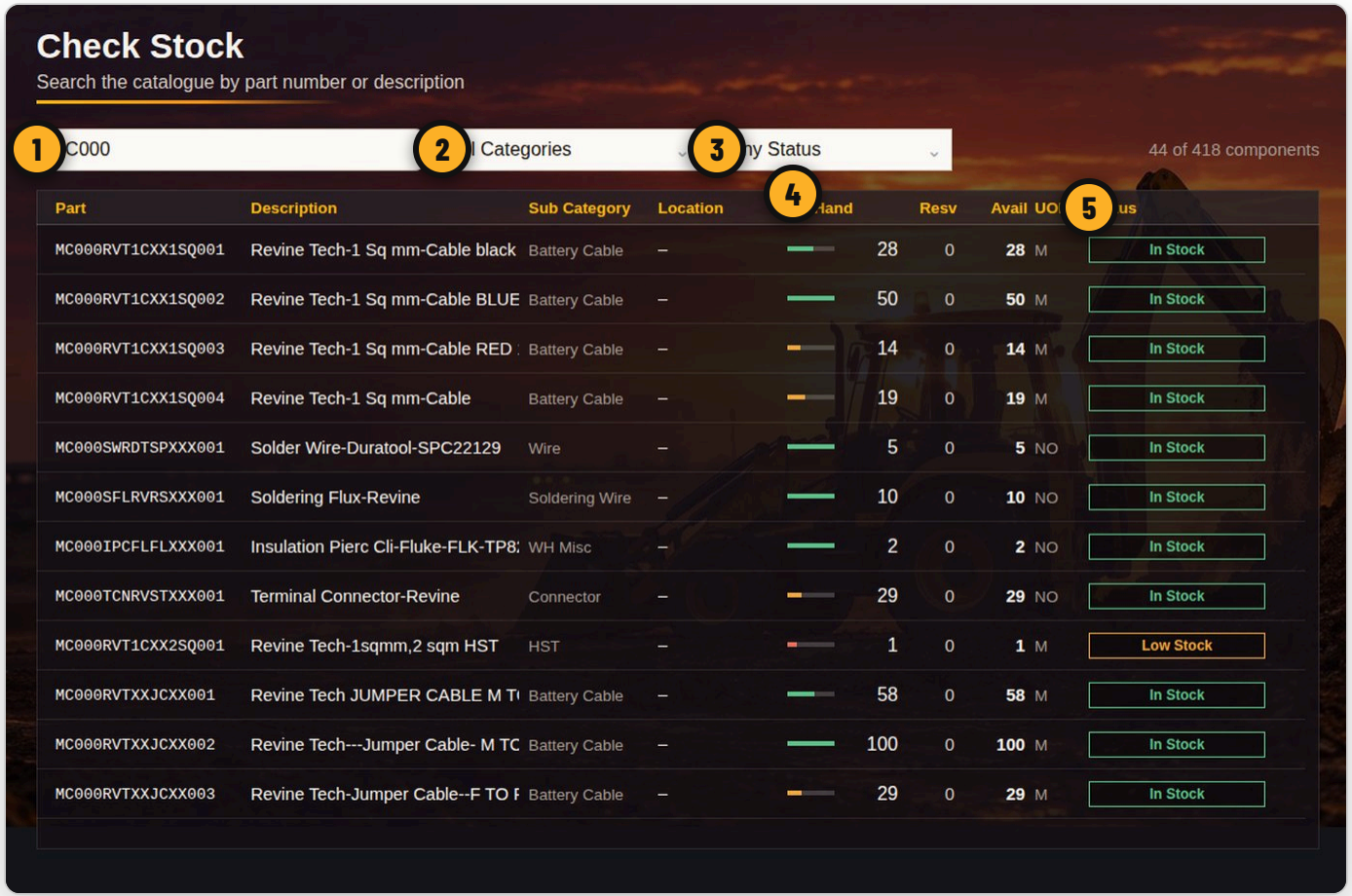


CLICK → WHAT HAPPENS

1. **Greeting**
   Good morning plus the person's name and role.
2. **Tiles**
   My open requests, parts in catalogue, low stock, out of stock.
3. **Your Requests**
   Each row opens the request detail. Open All lists everything.
4. **Menu**
   Only the pages an engineer needs.
5. **Live · Synced**
   Data refreshes itself; click to refresh now.
6. **Log Out**
   Back to the login page.

> **RESULT**
> An engineer sees their status at a glance.

CHECK STOCK

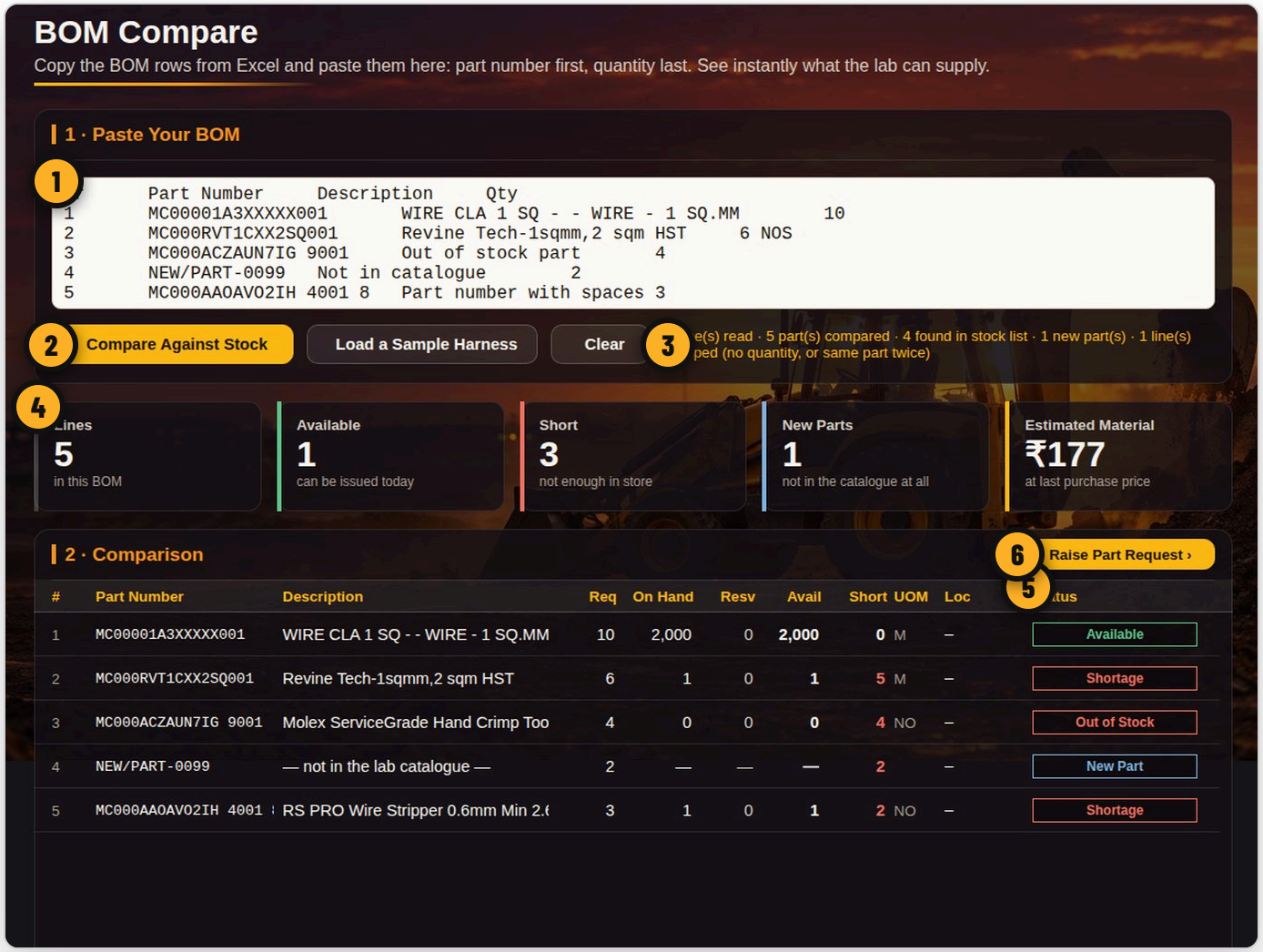


CLICK → WHAT HAPPENS

1. **Search**
   Part number, description or store location; results as you type.
2. **Category filter**
   Wires, connectors, relays and so on.
3. **Status filter**
   In stock, low, out of stock.
4. **On hand · Reserved · Available**
   Available = on hand minus what is reserved for requests.
5. **Status pill**
   Green in stock, amber low, red out of stock.

RESULT
Live stock for any of the 418 parts in seconds.

BOM COMPARE

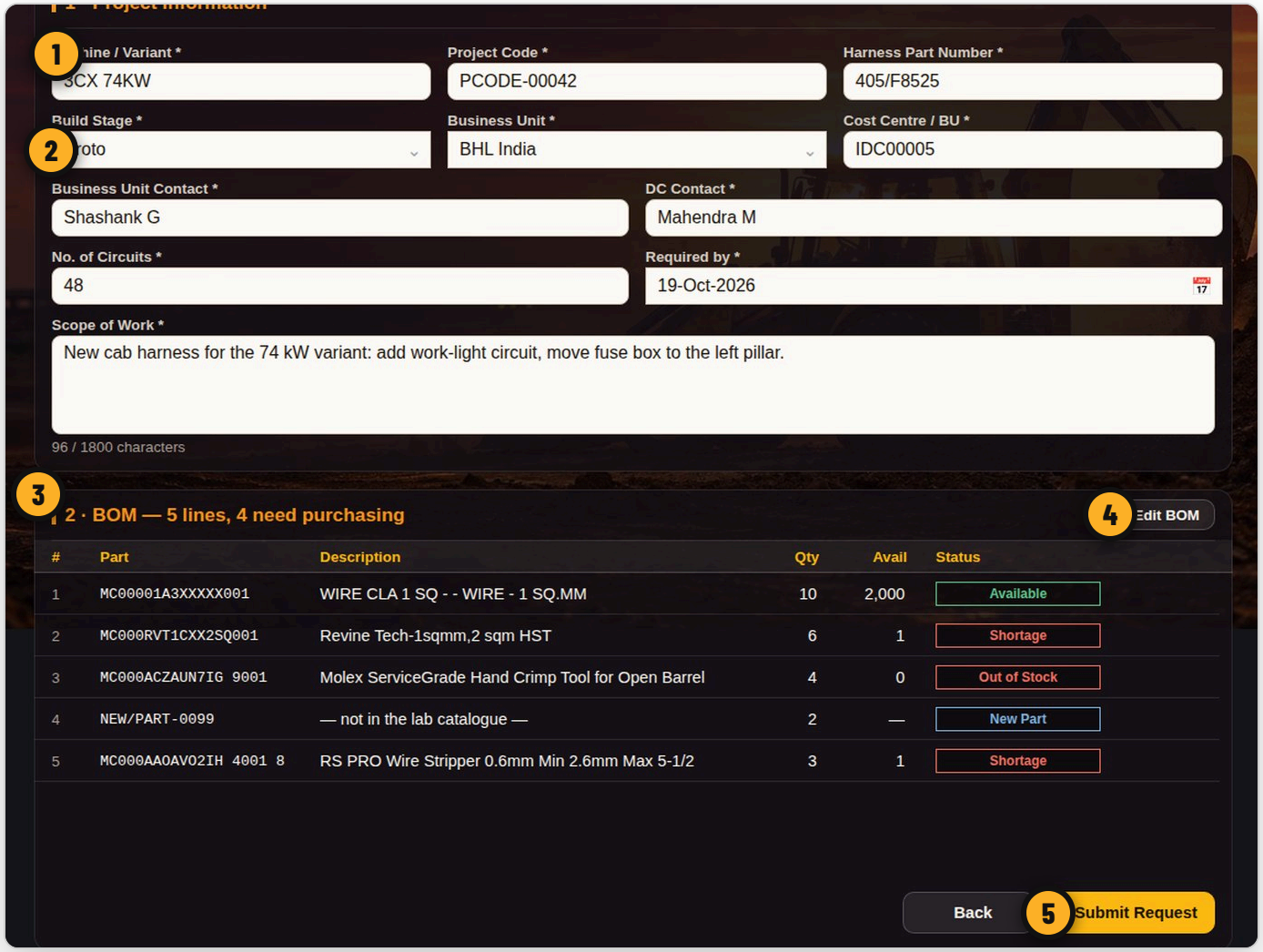


CLICK → WHAT HAPPENS

1. **Paste the BOM**
   Copy the rows straight from the Excel BOM: Sr, part number, description, qty. Units like NOS are fine.
2. **Compare Against Stock**
   The app re-reads the latest stock and checks every line of the BOM.
3. **Summary line**
   How many lines were read, parts compared, found in stock and new parts.
4. **Result tiles**
   Available today, short, new parts and the estimated material value.
5. **Status per line**
   Available · Shortage · Out of Stock · New Part, with on hand, reserved and short.
6. **Raise Part Request ›**
   Opens New Request with these lines already filled in.

RESULT
In seconds the engineer knows what the store can supply and what must be bought.

NEW REQUEST

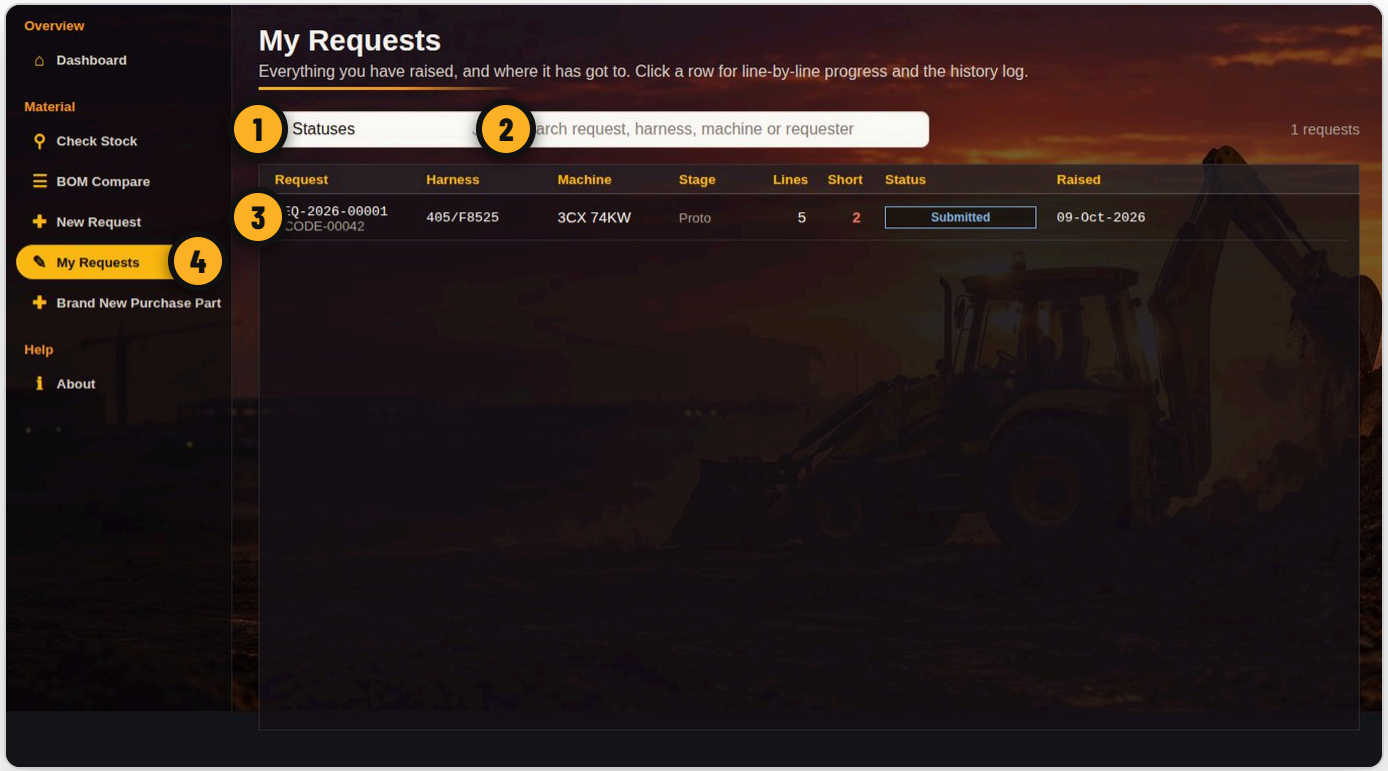


CLICK → WHAT HAPPENS

1. **Project information**
   Machine, project code, harness, stage, BU, cost centre, contacts, circuits, scope.
2. **Drop-downs**
   Stage and business unit are picked, not typed, so reports stay clean.
3. **BOM lines**
   Carried over from BOM Compare; the title says how many need purchasing.
4. **Edit BOM**
   Back to BOM Compare to change lines.
5. **Submit Request**
   Mandatory fields checked, request number REQ-YYYY-NNNNN created, lines saved.

RESULT
Request saved with all lines; green confirmation and the engineer lands on My Requests.

MY REQUESTS



CLICK → WHAT HAPPENS

1. **Status filter**
   Submitted, approved, ready, released, completed.

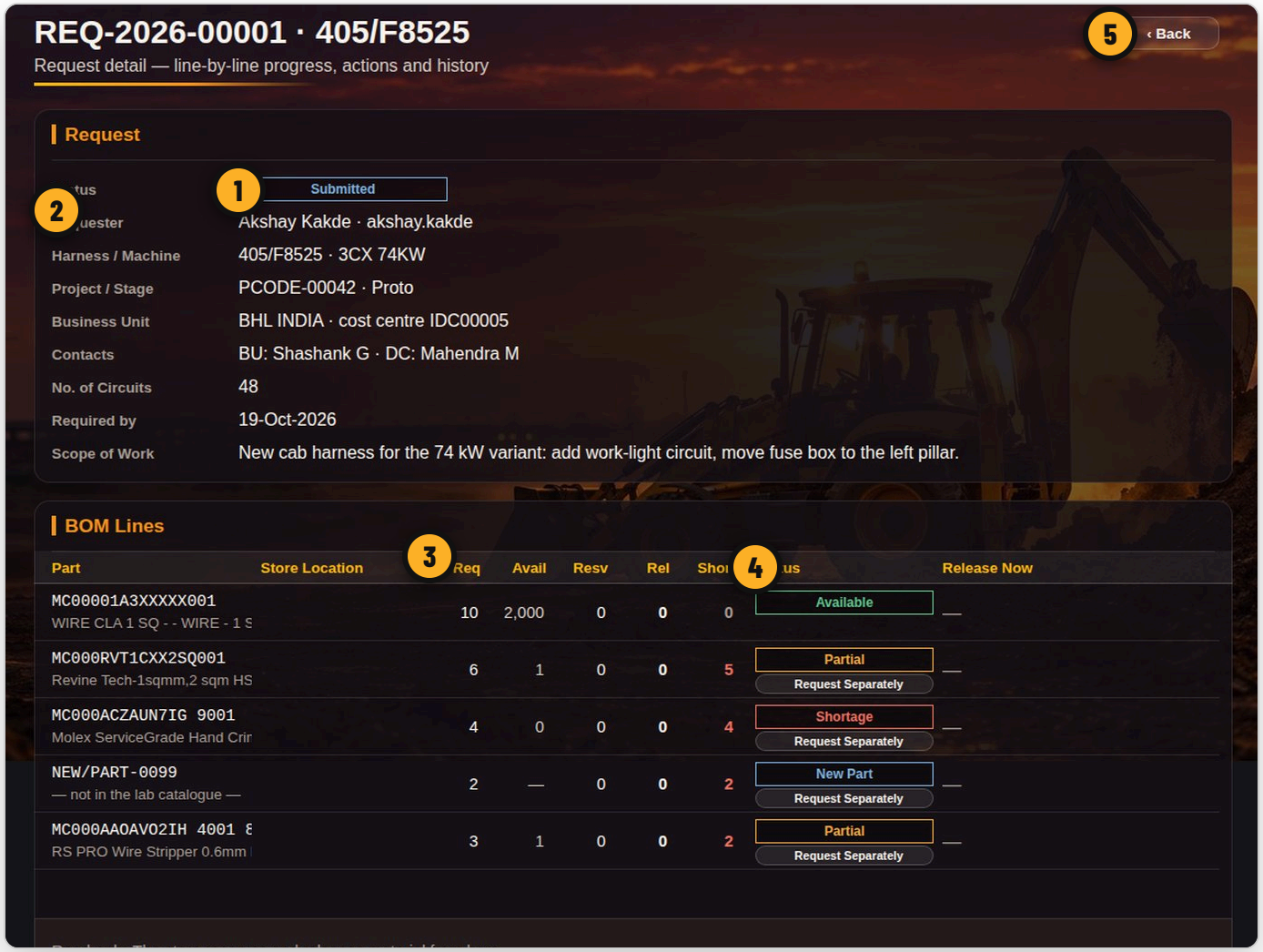
2. **Search**
   Request, harness, machine or requester.

3. **The new request**
   At the top, status Submitted, with lines and short count. Click → full detail.

4. **Badge**
   How many of my requests are still open.

RESULT
The engineer can track every request without asking the store.

REQUEST DETAIL · ENGINEER VIEW



CLICK → WHAT HAPPENS

1. **Status**
   Where the request is now: Submitted, Approved, Ready, Released.
2. **Header facts**
   Requester, harness, project, BU, contacts, required-by date, scope.
3. **Line by line**
   Requested, available, reserved, released and short per part.
4. **Line status**
   Available, Partial, Shortage or New Part.
5. **Back**
   Returns to My Requests.

RESULT
Full transparency for the requester: no calls to the store needed.

LAB ADMIN DASHBOARD

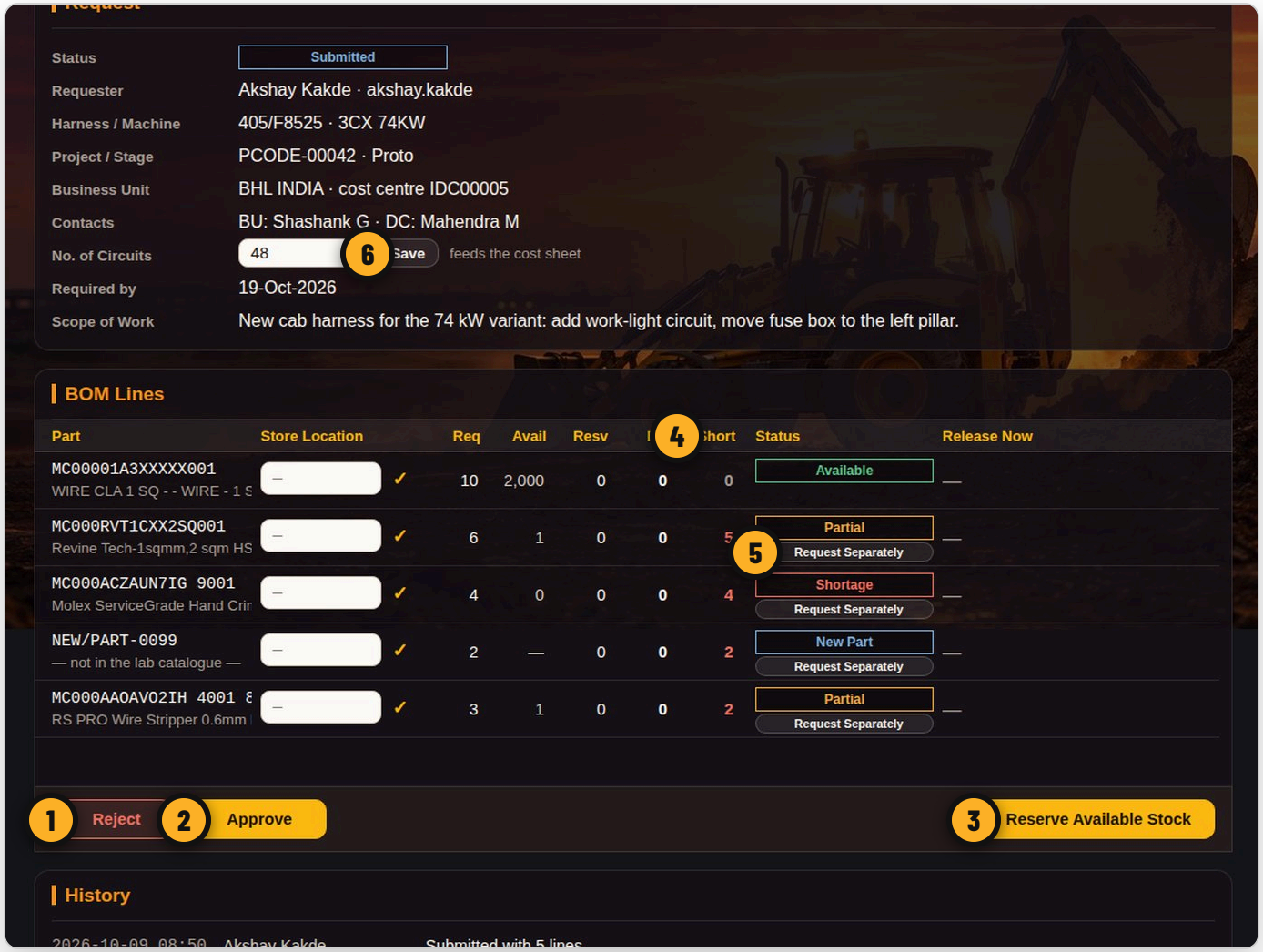


CLICK → WHAT HAPPENS

1. **Store tiles**
   Open requests, ready to release, purchase required, out of stock, low stock.
2. **Purchase and money**
   PRs and POs in progress, licenses needing attention, stock value.
3. **Needs Your Action**
   The engineer's new request is here the moment it is saved.
4. **Request Queue badge**
   Shows 1: one request waiting.
5. **View as**
   People with two roles switch view here.

> **RESULT**
> The store sees new work the moment it is submitted.

REQUEST DETAIL · LAB ADMIN

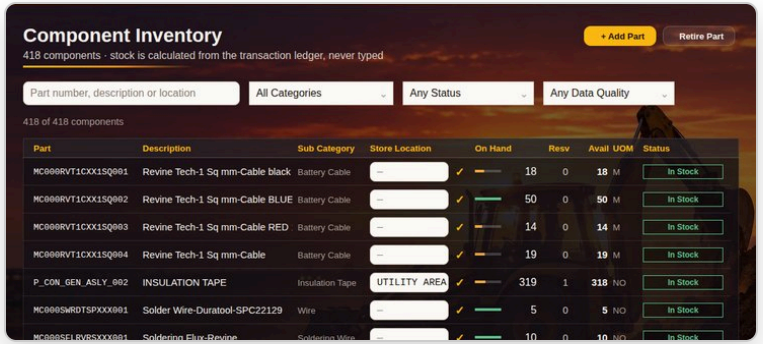


CLICK → WHAT HAPPENS

1. **Reject**
   Closes it with a reason in the history; the engineer still sees the record.
2. **Approve**
   Moves to Approved; the store can now reserve and release.
3. **Reserve Available Stock**
   Holds what is free for this request; what is short goes to Ongoing Purchase.
4. **Short column**
   What the store cannot supply today.
5. **Request Separately**
   Splits a shortfall into its own request so the rest can be released.
6. **Save circuits**
   Lab Admin corrects the circuit count used for the cost.

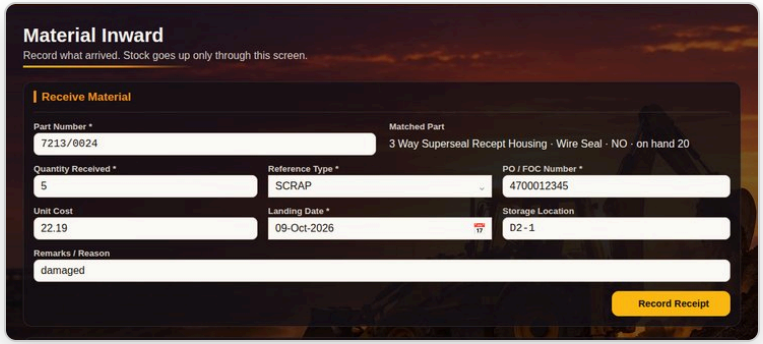
RESULT
The Lab Admin can act immediately; shortfalls go to the purchase list automatically.

OTHER PAGES (1 OF 2)



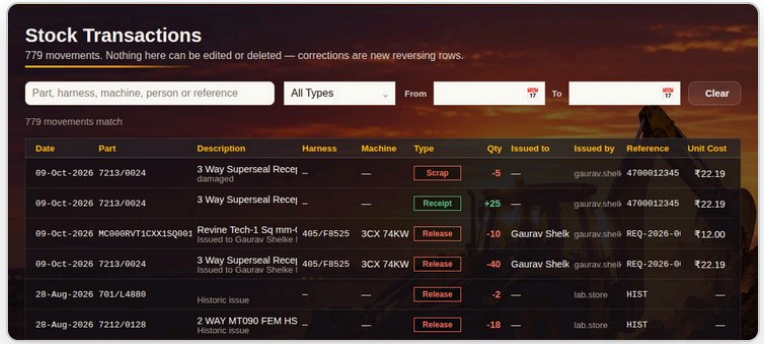
COMPONENT INVENTORY

Full catalogue with store locations; Lab Admin edits locations.



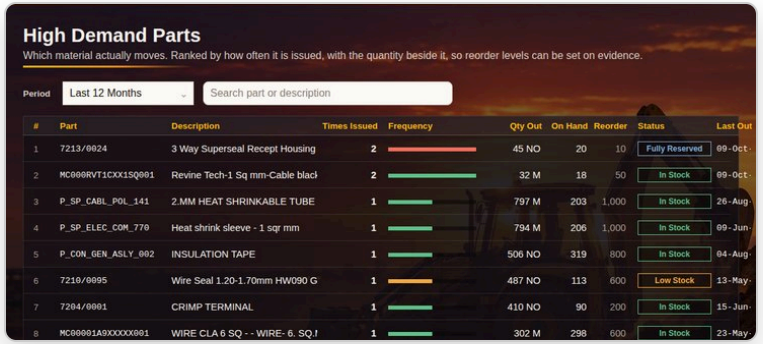
MATERIAL INWARD

Book goods received (PO, FOC, return, scrap); stock updates at once.



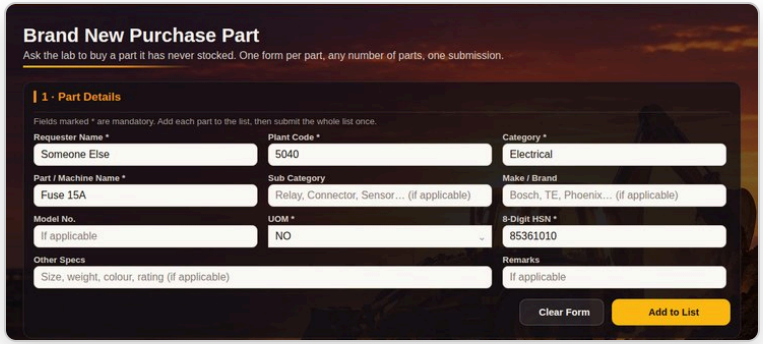
STOCK TRANSACTIONS

Every movement with date, reference and who issued it.



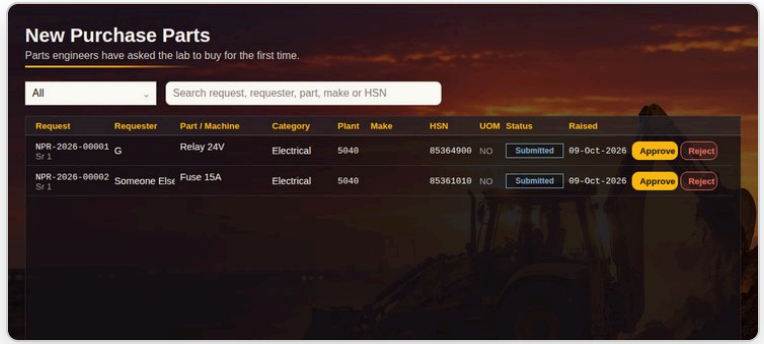
HIGH DEMAND PARTS

Most-issued parts for the chosen period, to plan buying.



BRAND NEW PURCHASE PART

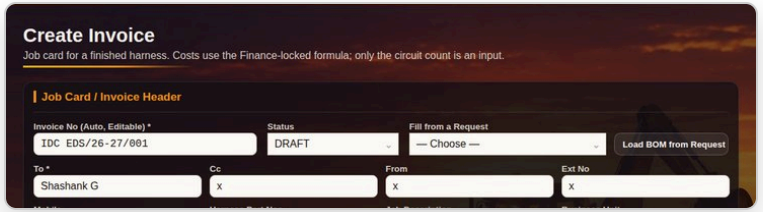
Engineer asks the lab to buy a part it has never stocked.



NEW PURCHASE PARTS

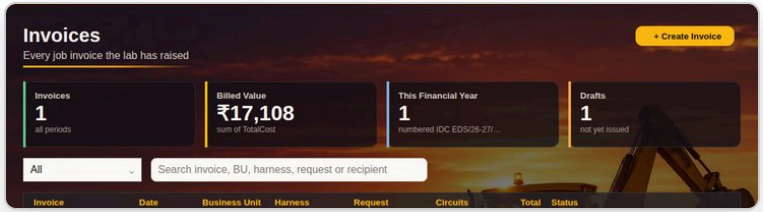
Lab Admin and Manager approve or reject those requests.

OTHER PAGES (2 OF 2)



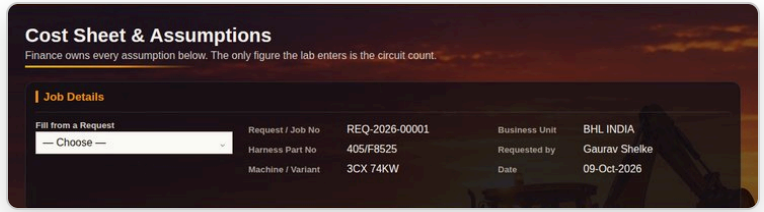
CREATE INVOICE

Job card for a released request; finance-locked cost formula.



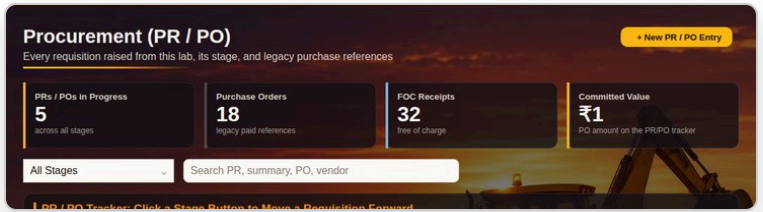
INVOICES

Every invoice raised; open, edit or issue.



COST SHEET

Quote a harness job: paste part, qty, cost.



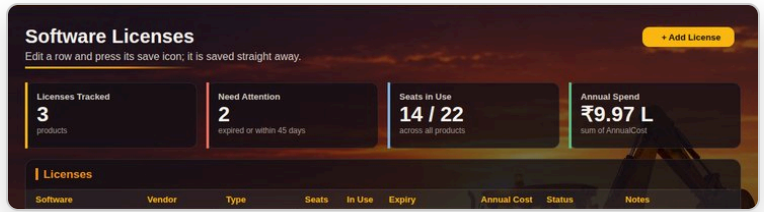
PROCUREMENT (PR/PO)

PR to PO to delivery, one click per stage.



ONGOING PURCHASE

All shortfalls in one list; what is on order.



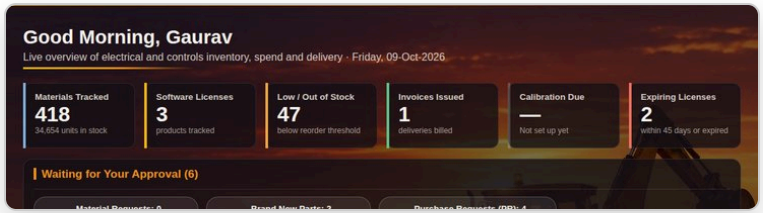
SOFTWARE LICENSES

Seats, renewal dates and yearly cost.



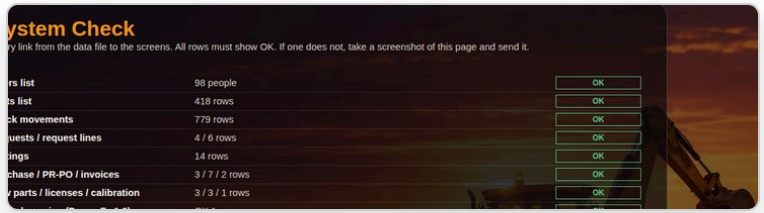
TEAM ACCESS

Roles, new people, password reset.



MANAGER DASHBOARD

Approvals, revenue, purchases, projects.



SYSTEM CHECK

18 health checks of the data links.

NEXT STEPS

PLAN

- Phase 1 pilot: 3 engineers + Lab Admin, 2 weeks.
- Phase 2: reserve → release → invoice live; manager dashboard.
- Later: move the data from Excel to a SharePoint list for more users and rows.

ASK

- Approval to start the Phase 1 pilot.
- Names of the 3 pilot engineers.
- Review meeting at the end of the 2 weeks.

Appendix: one slide for every page of the app, for questions.

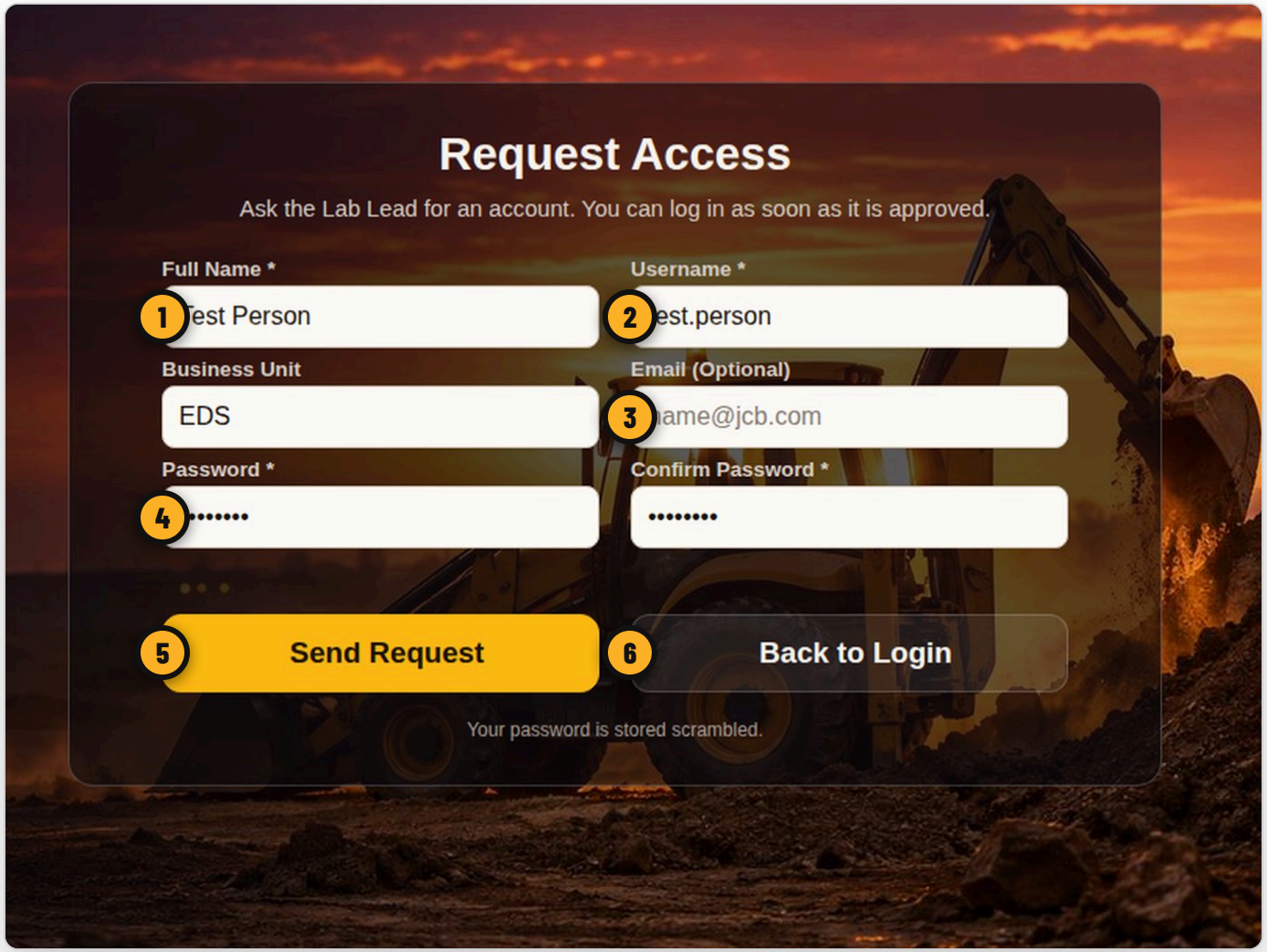
APPENDIX

EVERY PAGE IN DETAIL

One slide per page, for questions. Click a page to jump; Home returns to the start.

18 Register	Login	
19 Set or reset password	Login	
20 System Check	Support	
21 Request Queue	Lab Admin	
22 Component Inventory	Lab Admin	
23 Material Inward	Lab Admin	
24 Stock Transactions	Lab Admin	
25 High Demand Parts	Lab Admin	
26 Brand New Purchase Part	Engineer	
27 New Purchase Parts	Lab Admin / Manager	
28 Create Invoice	Lab Admin	
29 Invoices	Lab Admin	
30 Cost Sheet	Lab Admin	
31 Procurement (PR / PO)	Lab Admin	
32 Ongoing Purchase	Lab Admin	
33 Software Licenses	Lab Admin	
34 Team Access	Lab Admin / Lab Lead	
35 Manager Dashboard	Manager	
36 About · Data Health	Lab Admin	
37 Print / PDF	Lab Admin	

REGISTER

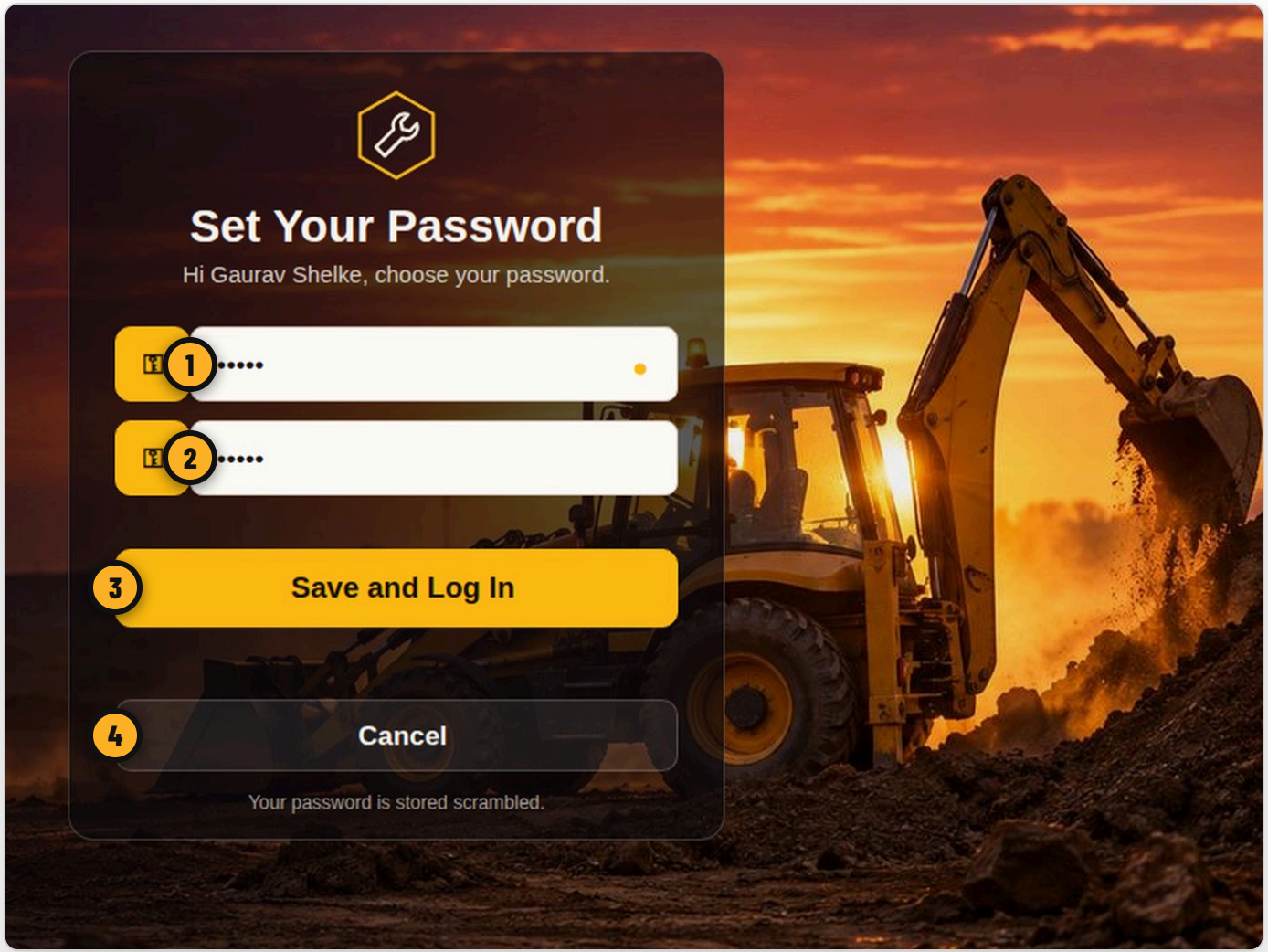


CLICK → WHAT HAPPENS

1. **Full Name**
   Shown on requests and the dashboard greeting.
2. **Username**
   What the person logs in with.
3. **Business Unit · Email**
   Optional; used on reports.
4. **Password + confirm**
   At least 4 characters.
5. **Send Request**
   Adds the person with role Pending; the Lab Admin sets the role in Team Access.
6. **Back to Login**
   Cancels.

> **RESULT**
> New people get in without a form or a mail chain.

SET OR RESET PASSWORD

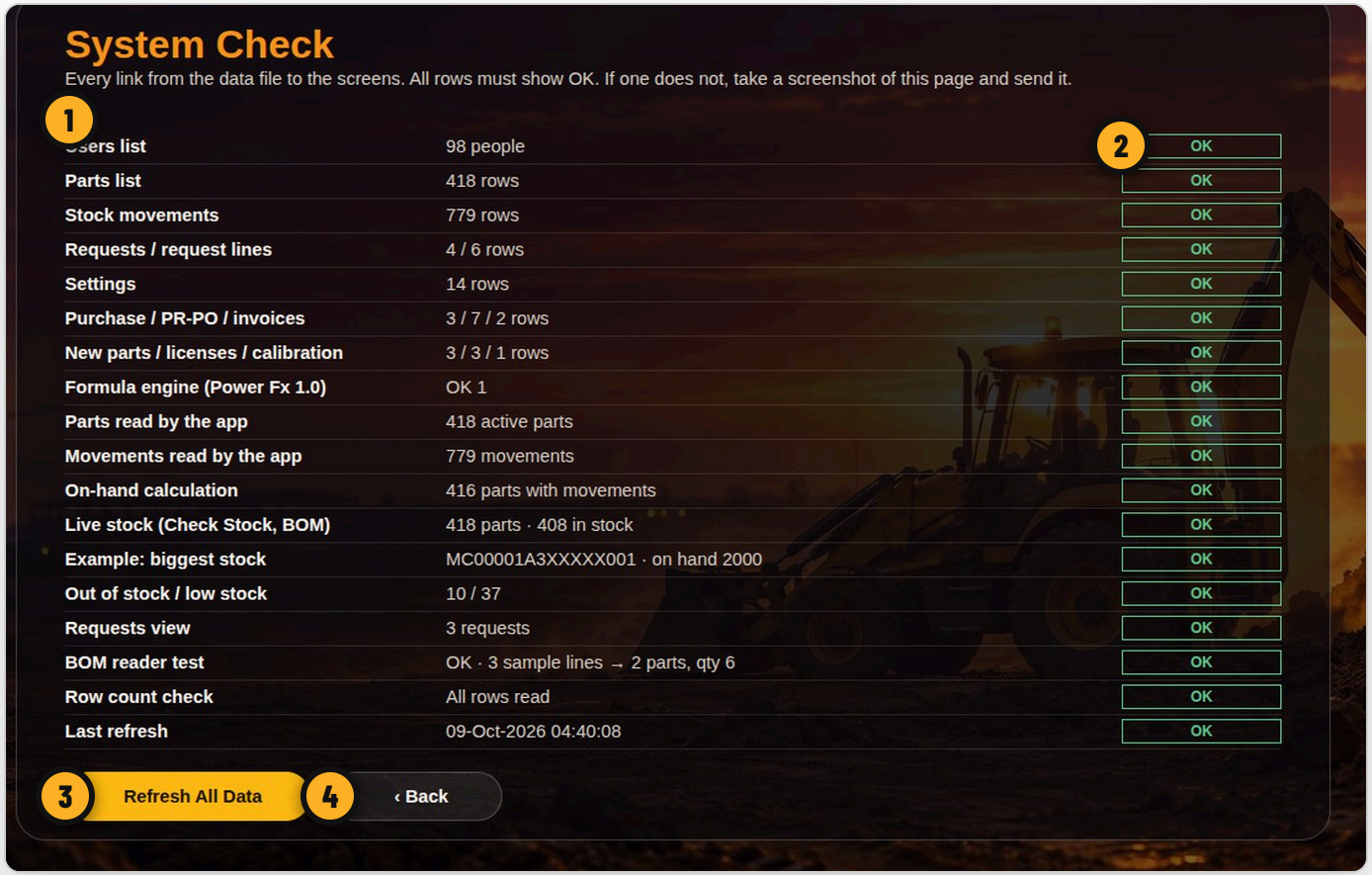


CLICK → WHAT HAPPENS

1. **New password**
   At least 4 characters.
2. **Confirm**
   Must match.
3. **Save and Log In**
   Password saved scrambled; the person goes straight in.
4. **Cancel**
   Back to the login box.

RESULT
Forgot Password and first login both end here. No waiting for an admin.

SYSTEM CHECK

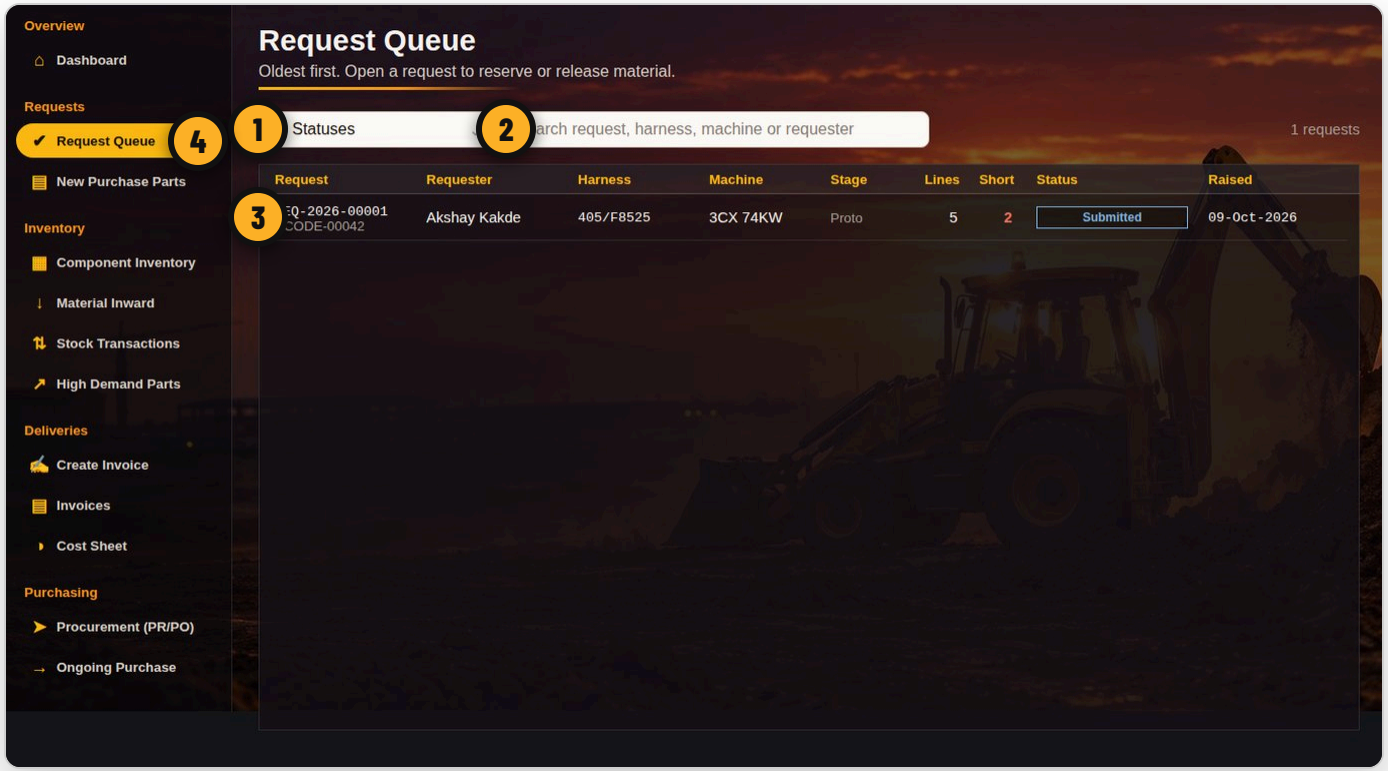


CLICK → WHAT HAPPENS

1. **18 checks**
   Every table connected and read, stock calculation, BOM reader test, last refresh.
2. **OK per row**
   Any row not OK points to the exact problem.
3. **Refresh All Data**
   Re-reads every table now.
4. **Back**
   Returns to the login page.

RESULT
One screenshot tells support what is wrong.

REQUEST QUEUE

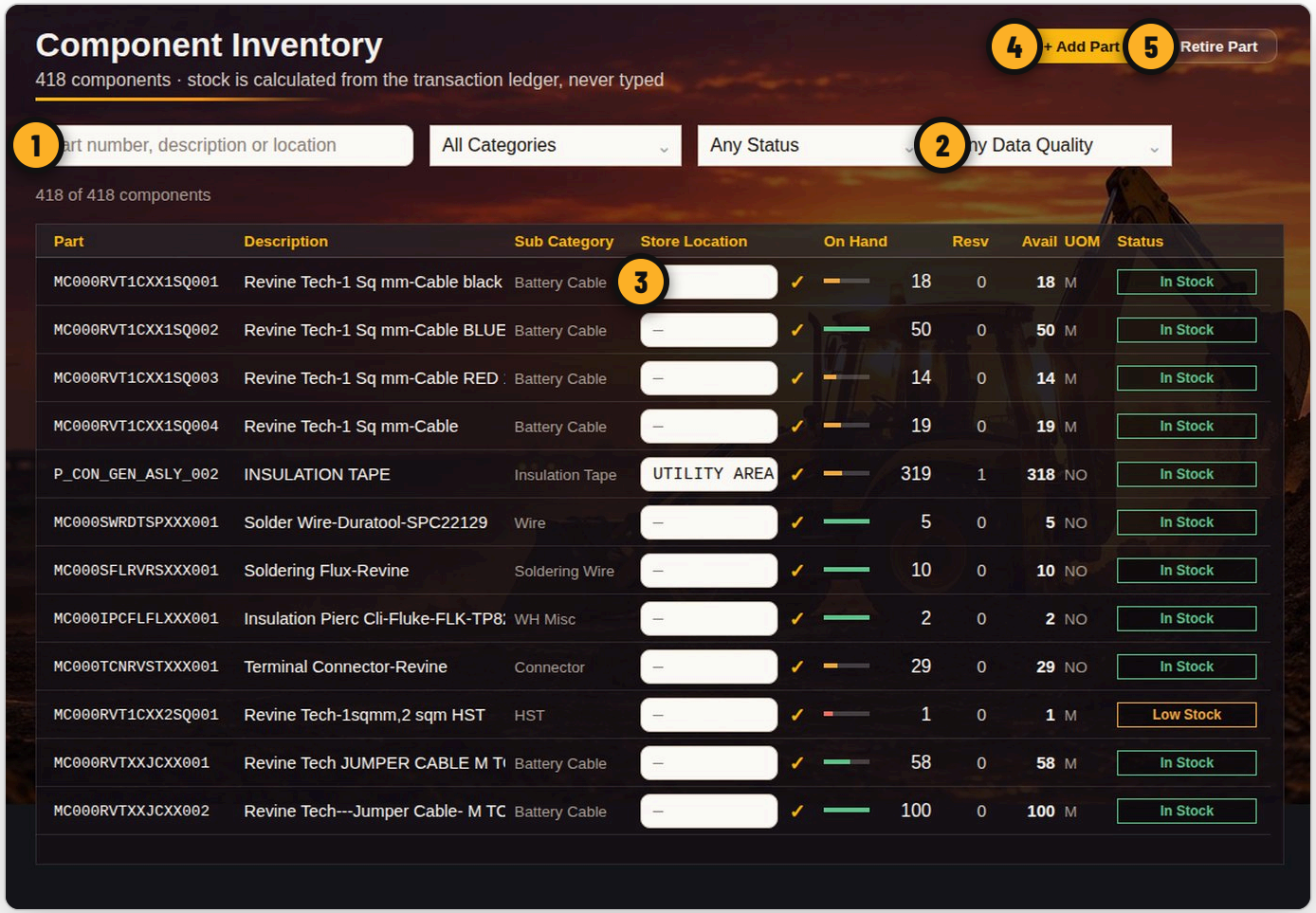


CLICK → WHAT HAPPENS

1. **Status filter**
   Submitted, approved, ready for release, purchase required…
2. **Search**
   Request, harness, machine or requester.
3. **Request row**
   Lines, short count, status, date. Click → request detail.
4. **Badge**
   Requests waiting for action.

> **RESULT**
> The store's to-do list, oldest first.

COMPONENT INVENTORY

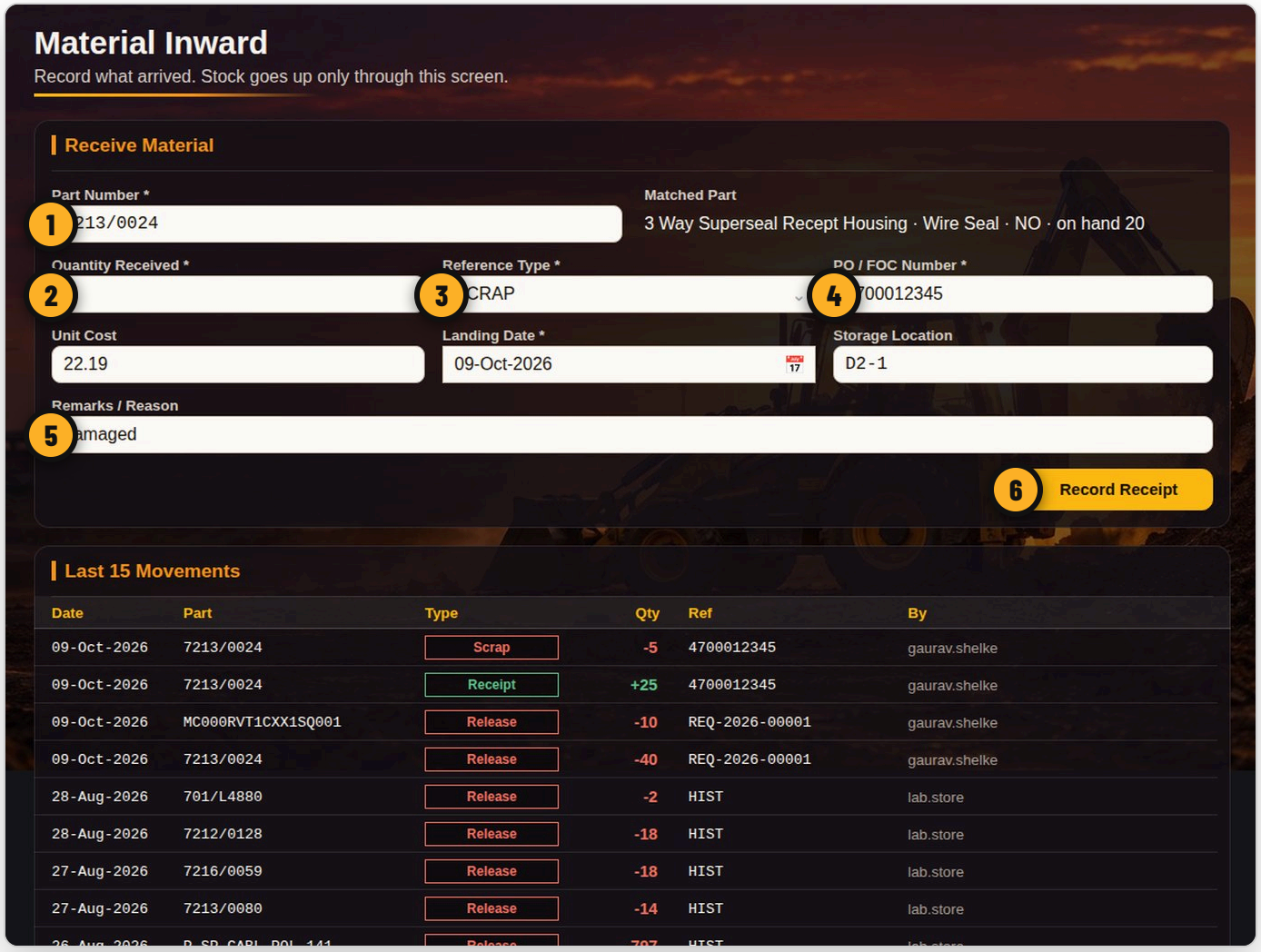


CLICK → WHAT HAPPENS

1. **Search**
   Part, description or location.
2. **Data quality filter**
   Finds parts with missing or odd data in the sheet.
3. **Location**
   Lab Admin types the shelf location in the row; saved at once.
4. **+ Add Part**
   Part number, description, category, UOM, location, supplier, cost.
5. **Retire Part**
   Takes an obsolete part out of the active list.

RESULT
The catalogue stays correct without opening Excel.

MATERIAL INWARD

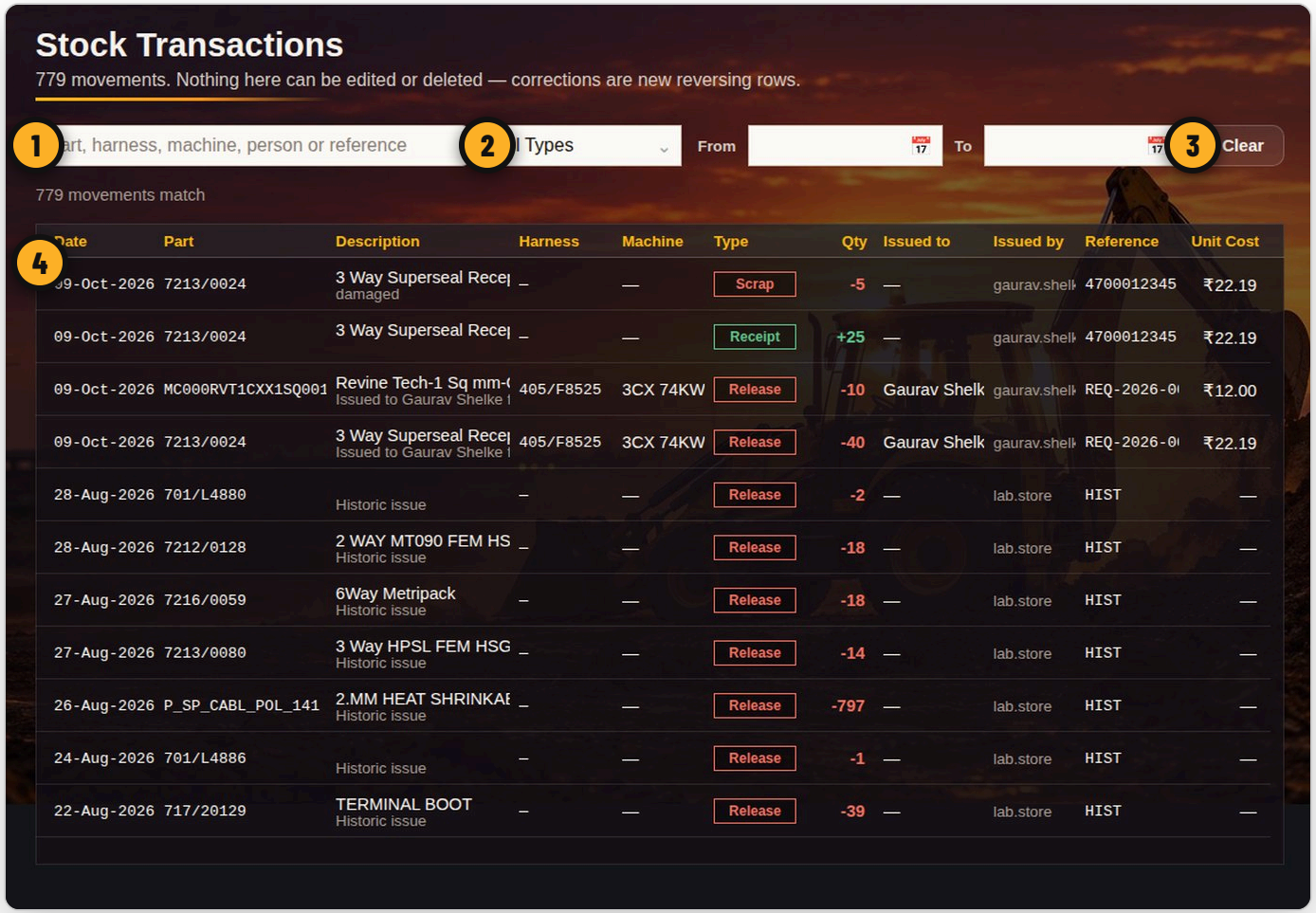


CLICK → WHAT HAPPENS

1. **Part number**
   Checked against the catalogue as you type.
2. **Quantity**
   Must be more than zero.
3. **Type**
   PO, FOC, Return, Adjust+, Adjust−, Scrap.
4. **Reference**
   PO number or note.
5. **Reason**
   Required for adjustments and scrap.
6. **Record Receipt**
   Adds a stock movement; on hand updates at once.

> **RESULT**
> Stock goes up only through this page, so every unit is traceable.

STOCK TRANSACTIONS



CLICK → WHAT HAPPENS

1. **Search**
   Part, harness, machine, person or reference.
2. **Type filter**
   Receipt, release, return, adjustments, scrap.
3. **Clear**
   Resets the filters.
4. **Ledger**
   Date, part, quantity, reference, unit cost and who did it.

RESULT
Full audit trail of every unit in and out.

HIGH DEMAND PARTS

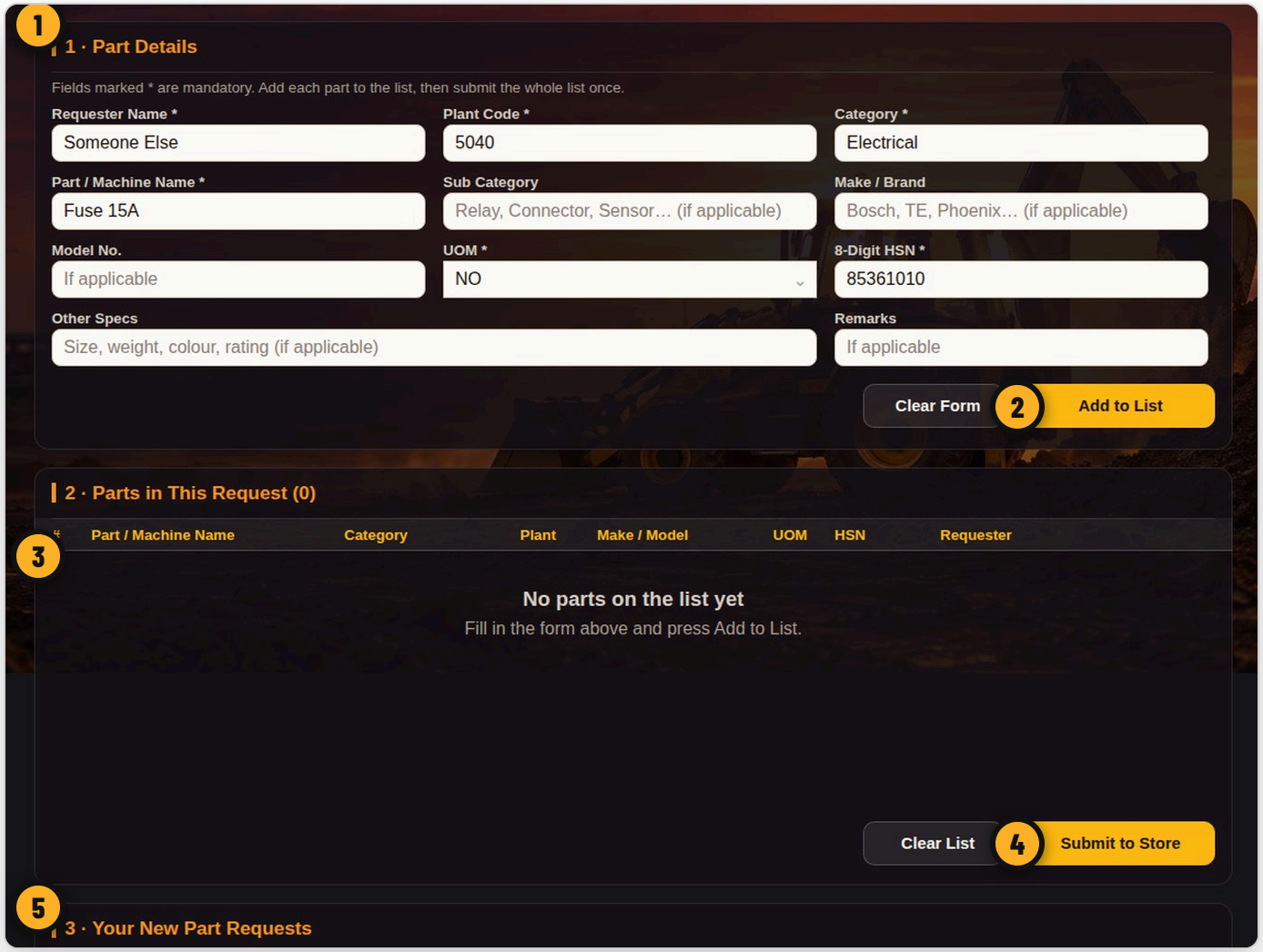


CLICK → WHAT HAPPENS

1. **Period**
   Last 90 days, 180 days, 12 months or everything.
2. **Search**
   Part or description.
3. **Ranking**
   Most-issued parts first, with stock status and last issue date.

> **RESULT**
> Buy ahead for the parts every build uses.

BRAND NEW PURCHASE PART

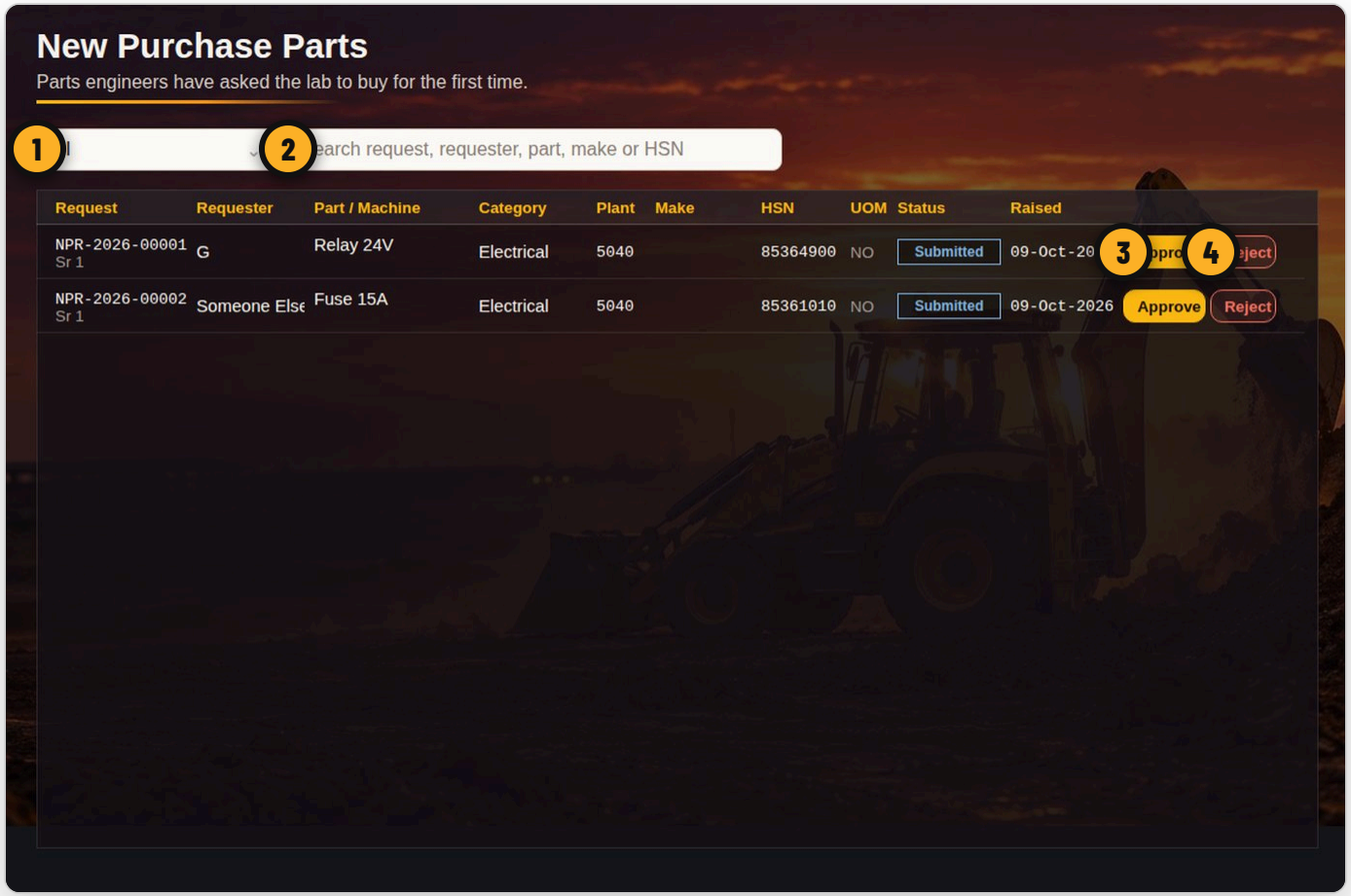


CLICK → WHAT HAPPENS

1. **Part details**
   Name, category, make, model, UOM, HSN, specs.
2. **Add to List**
   Several new parts in one request.
3. **Parts in this request**
   Edit or remove before sending.
4. **Submit to Store**
   Numbered request goes to the New Purchase Parts queue.
5. **Your New Part Requests**
   Status of everything you asked for.

> **RESULT**
> Engineers ask for parts the lab has never stocked, in a fixed format.

NEW PURCHASE PARTS

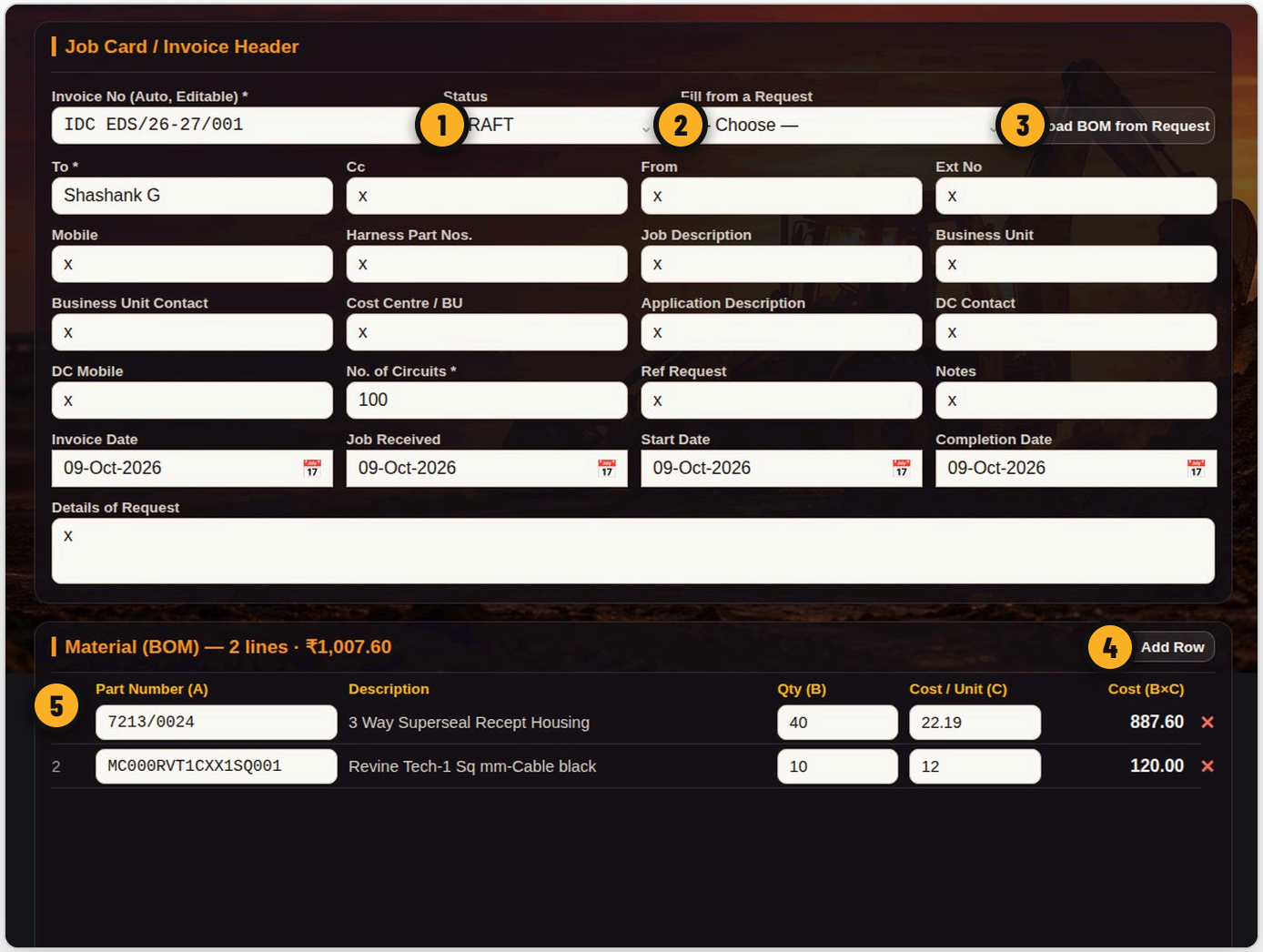


CLICK → WHAT HAPPENS

1. **Status filter**
   Submitted, approved, rejected, draft.
2. **Search**
   Request, requester, part, make or HSN.
3. **Approve**
   Marked approved; add it to the catalogue once it has a part number.
4. **Reject**
   Marked rejected; the engineer sees it.

> **RESULT**
> Every new-part request is decided in one place.

CREATE INVOICE

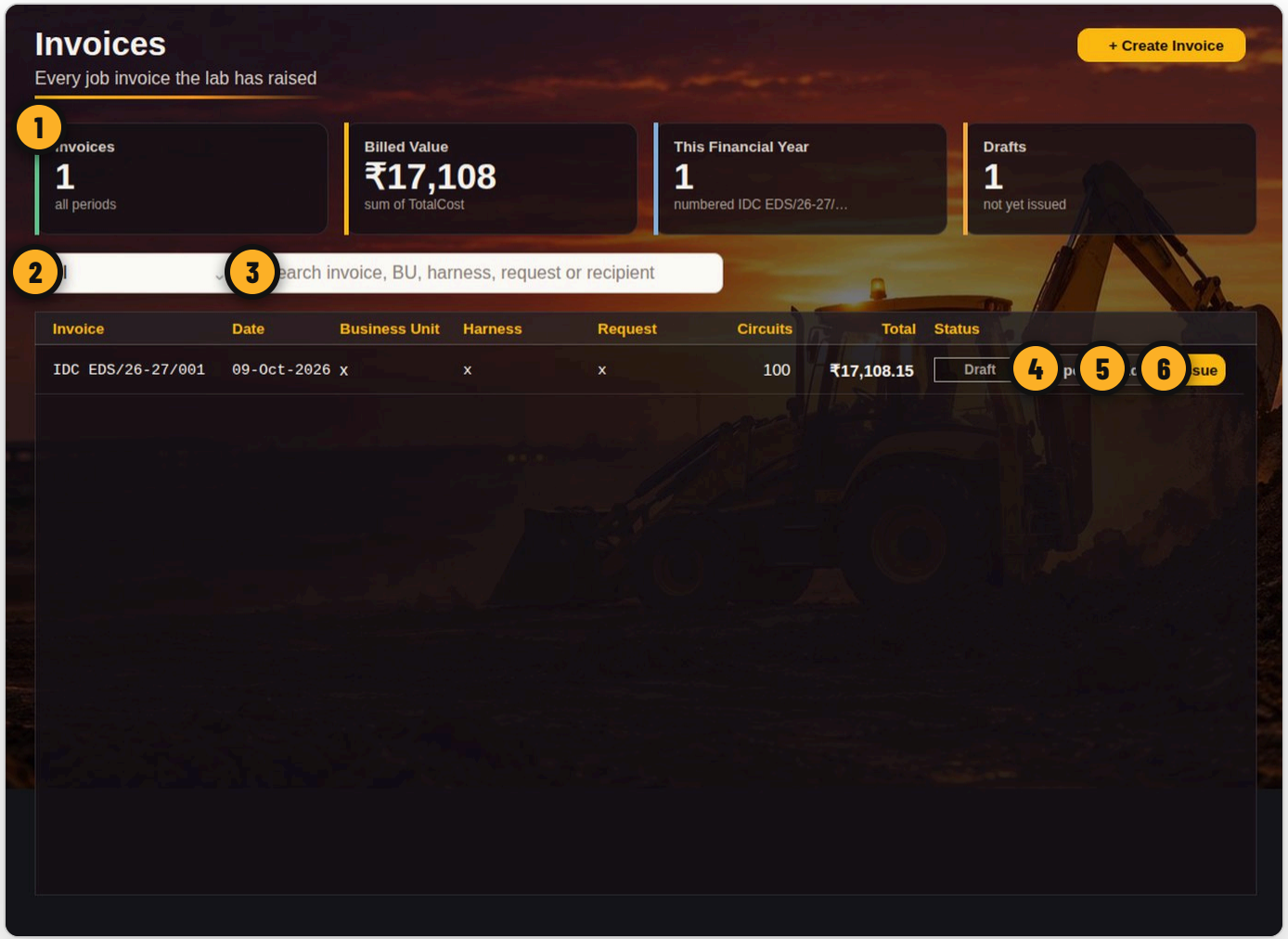


CLICK → WHAT HAPPENS

1. **Status**
   Draft until it is issued.
2. **Pick the request**
   Fills job details from a released request.
3. **Load BOM from Request**
   Material lines with qty and unit cost.
4. **+ Add Row**
   Extra material by hand.
5. **Material lines**
   Qty × cost per line, total in the title.
6. **Save Invoice · Print / PDF**
   At the bottom: saves with a unique number and prints the job card.

RESULT
Job invoice built from what was really released.

INVOICES

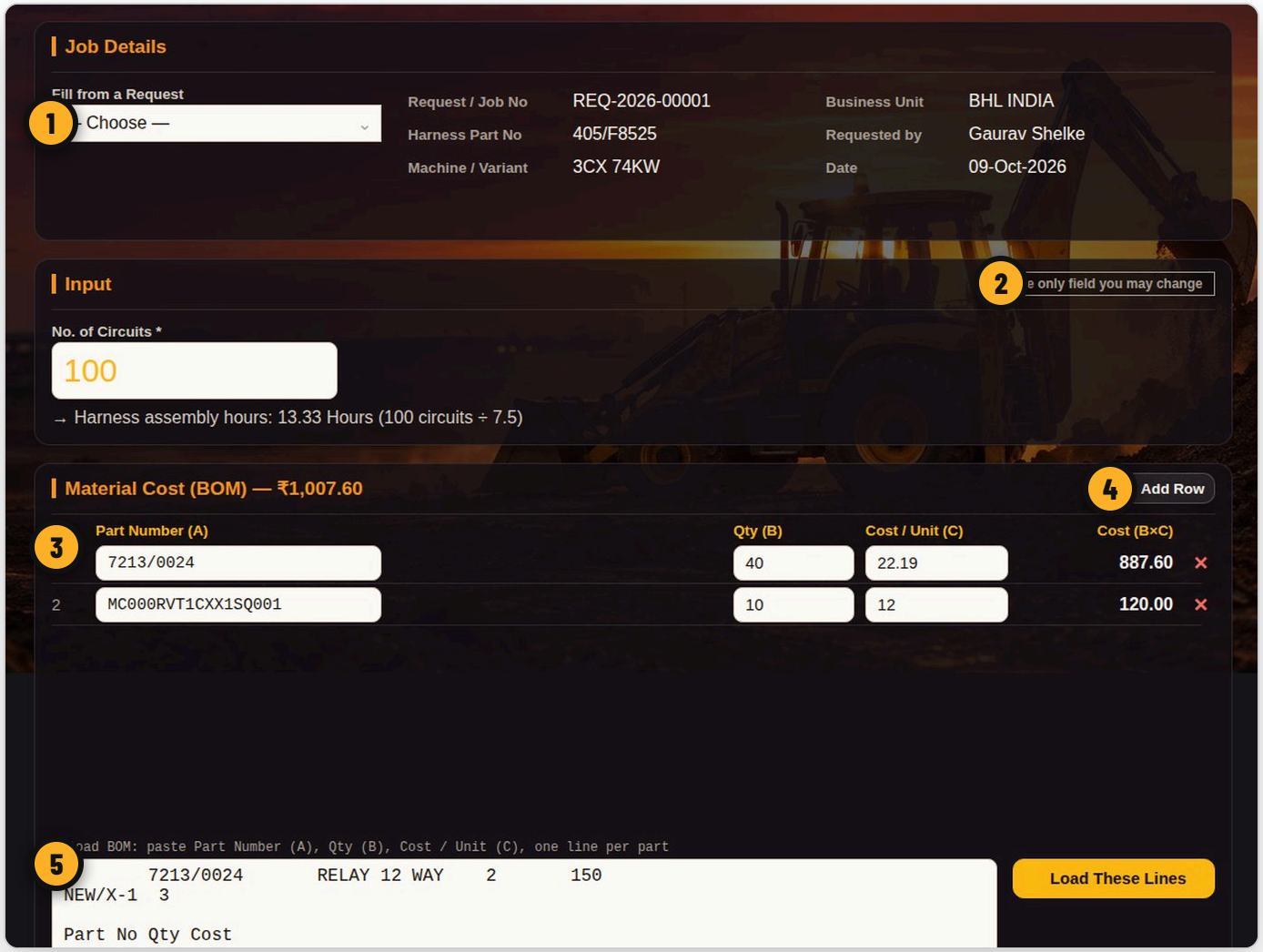


CLICK → WHAT HAPPENS

1. **Tiles**
   Invoices, billed value, this financial year, drafts.
2. **Status filter**
   Draft or issued.
3. **Search**
   Invoice, BU, harness, request or recipient.
4. **Open**
   Read view and print.
5. **Edit**
   Back to Create Invoice with its lines.
6. **Issue**
   Marks it issued.

> **RESULT**
> All lab invoices in one list.

COST SHEET

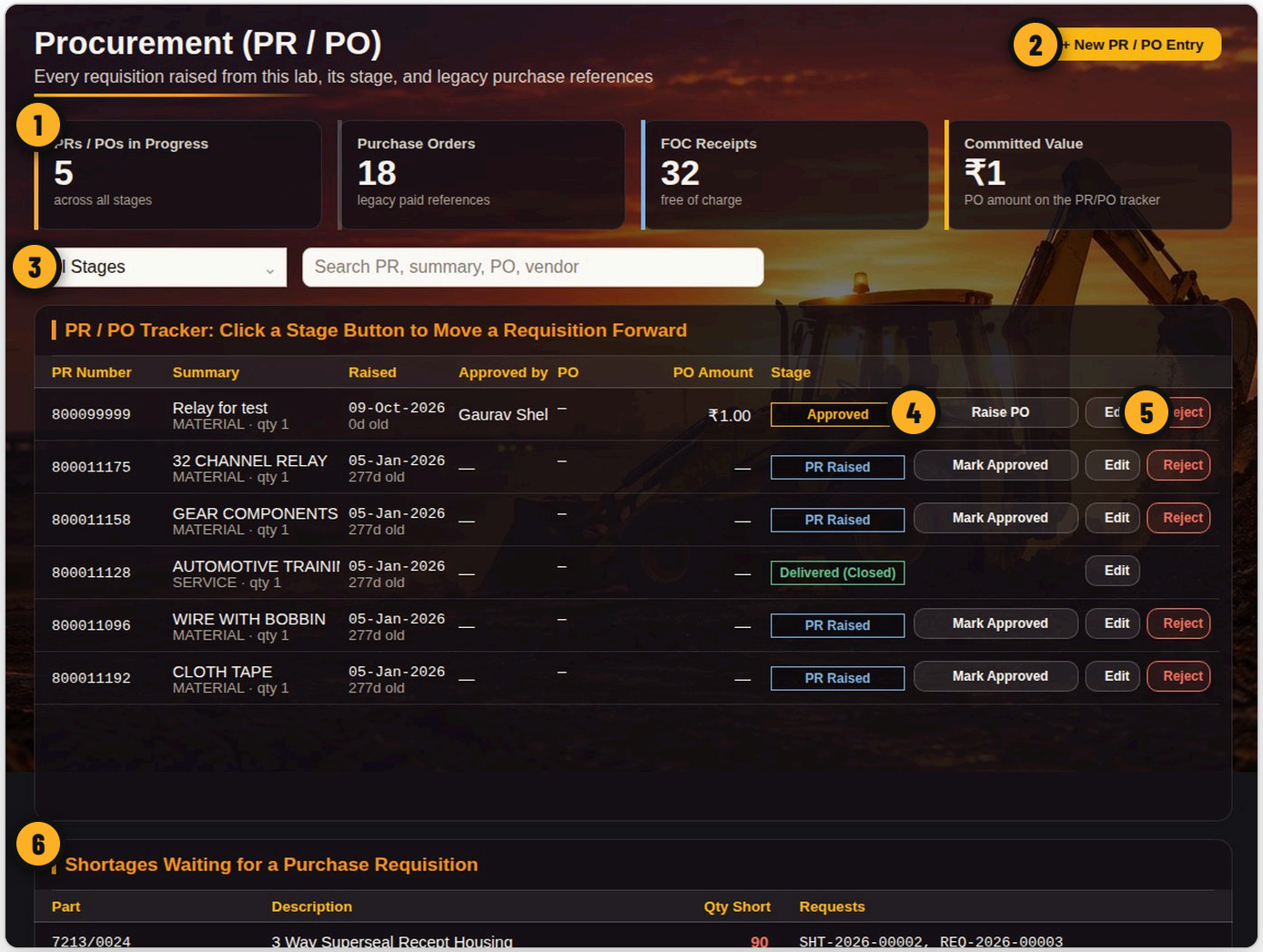


CLICK → WHAT HAPPENS

1. **Pick a job**
   Request or invoice to cost.
2. **Input**
   Number of circuits is the only field you change.
3. **Material cost**
   Part, qty, cost per unit, line cost.
4. **+ Add Row**
   Extra lines.
5. **Paste lines**
   Part, qty, cost straight from Excel; Load These Lines.
6. **Calculated · Print**
   Below: labour and overheads from the finance-locked assumptions; print both.

RESULT
A quote for a harness job in minutes.

PROCUREMENT (PR / PO)

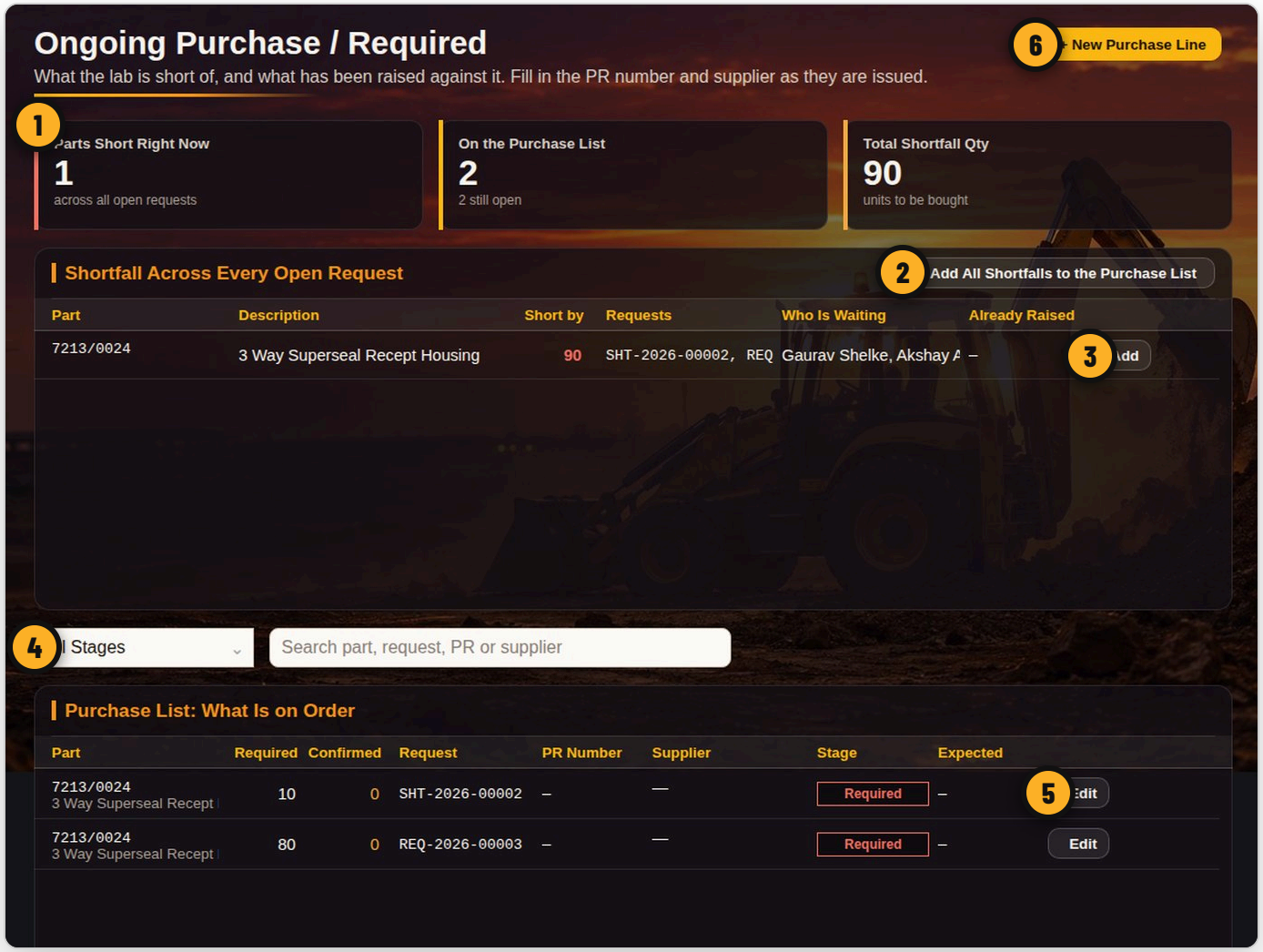


CLICK → WHAT HAPPENS

1. **Tiles**
   PRs/POs in progress, purchase orders, FOC receipts, committed value.
2. **+ New PR / PO Entry**
   PR number, summary, vendor, amount.
3. **Stage filter**
   PR raised, approved, PO raised, GRN, delivered, rejected.
4. **Stage button**
   One click moves it to the next stage.
5. **Reject**
   Closes the PR.
6. **Shortages waiting for a PR**
   From open requests, so nothing is forgotten.

RESULT
Every purchase tracked from PR to delivery.

ONGOING PURCHASE

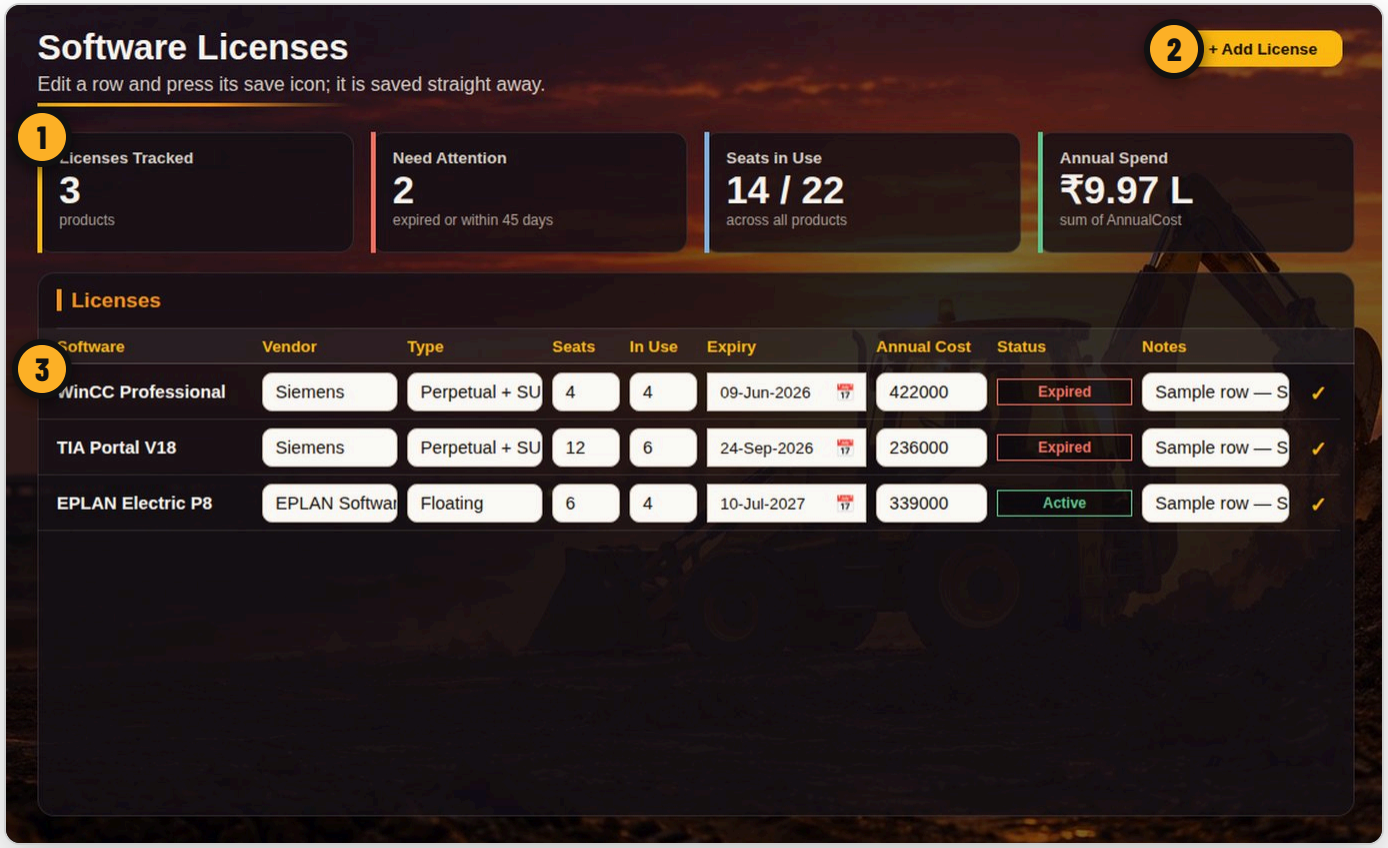


CLICK → WHAT HAPPENS

1. **Tiles**
   Parts short now, on the purchase list, total shortfall qty.
2. **Add All Shortfalls**
   Puts every open shortfall on the purchase list.
3. **Add**
   One part for one request.
4. **Stage filter**
   Required, PR raised, PO raised, in transit, received.
5. **Edit**
   Supplier, PR number, stage.
6. **+ New Purchase Line**
   Buy something not tied to a request.

> **RESULT**
> What is short across all requests, and what is already on order.

SOFTWARE LICENSES



CLICK → WHAT HAPPENS

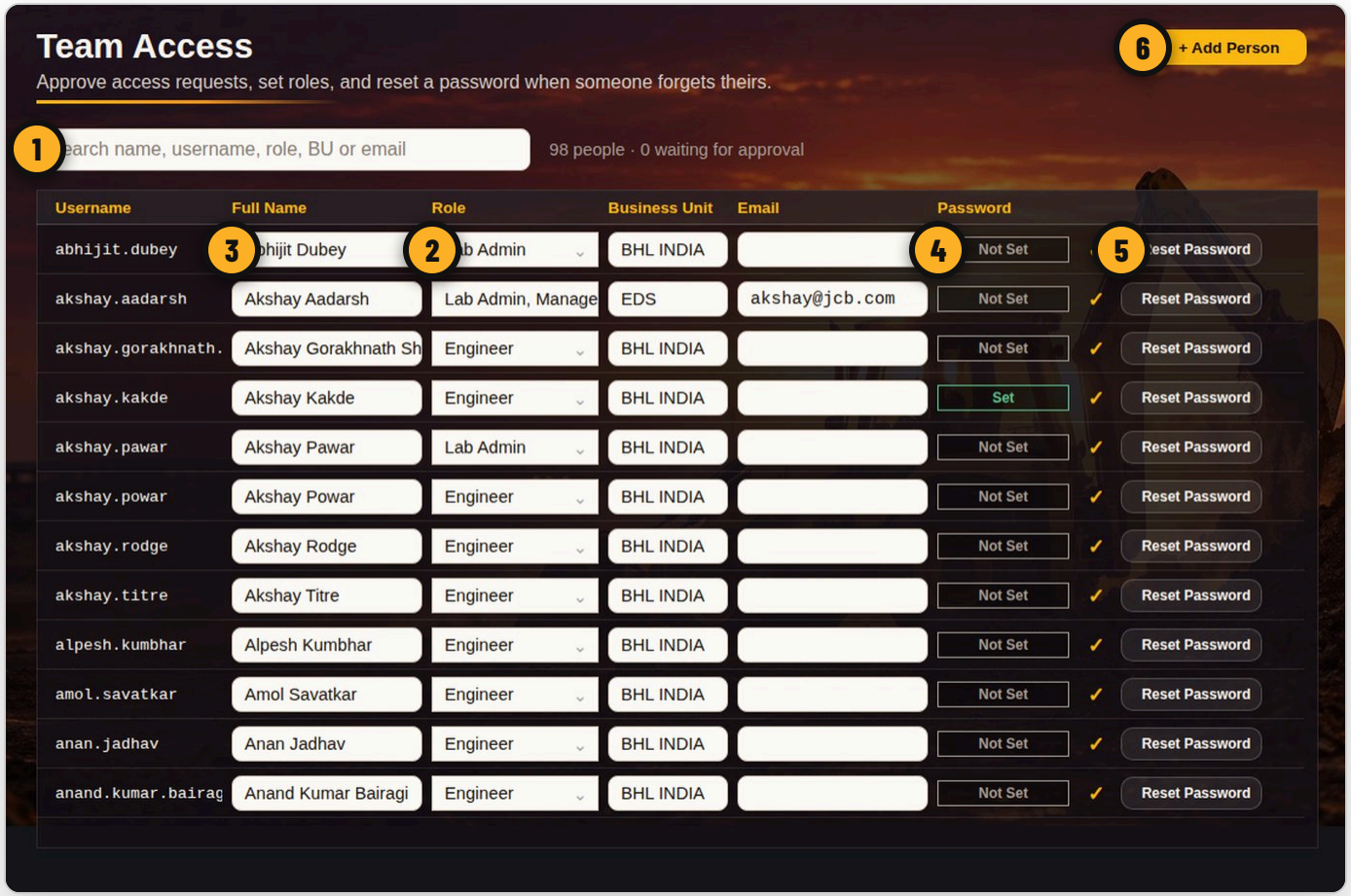
1. **Tiles**
   Licenses tracked, need attention, seats in use, annual spend.

2. **+ Add License**
   Name, vendor, seats, expiry, cost.

3. **List**
   Edit seats or dates in the row.

RESULT
No renewal is missed.

TEAM ACCESS

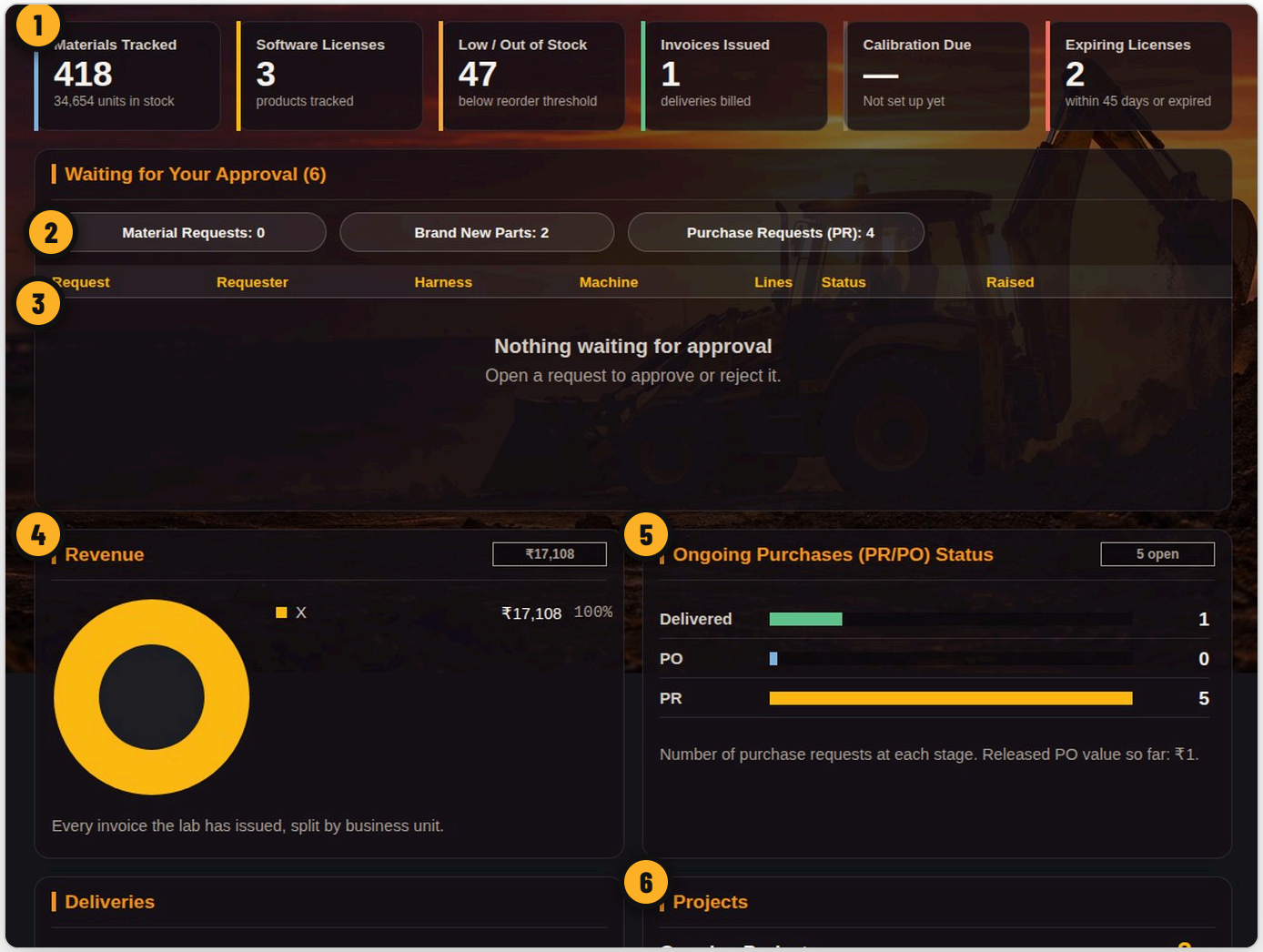


CLICK → WHAT HAPPENS

1. **Search**
   Name, username, role, BU or email.
2. **Role**
   Drop-down incl. combined roles like Engineer, Manager.
3. **Name · BU · Email**
   Edit in the row.
4. **Password**
   Set or Not Set; the password itself is never shown.
5. **Reset Password**
   Clears it; the person chooses a new one at next login.
6. **+ Add Person**
   Creates the user directly.

> **RESULT**
> Who can see what, managed without opening Excel.

MANAGER DASHBOARD

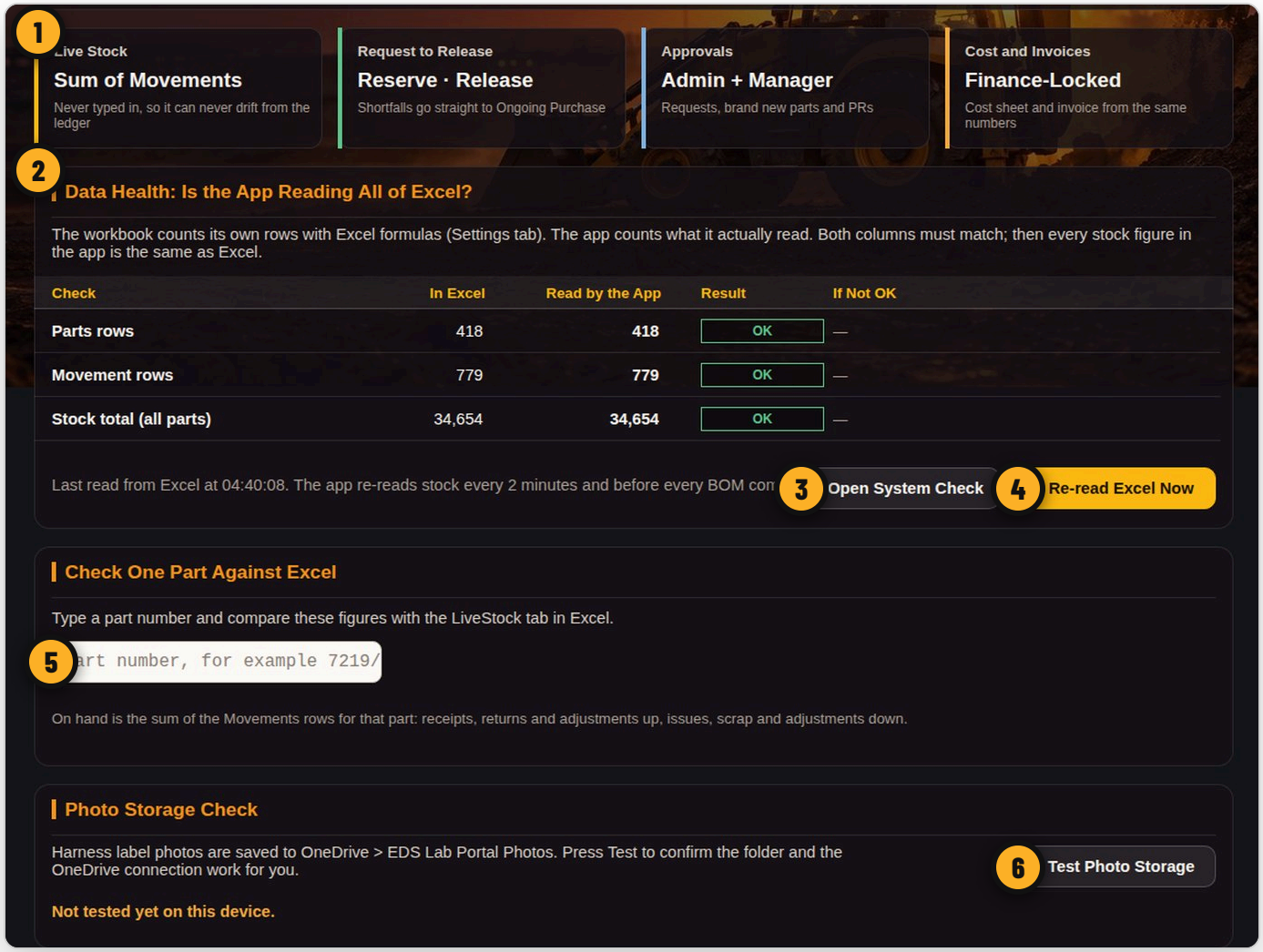


CLICK → WHAT HAPPENS

1. **Tiles**
   Materials, licenses, low / out of stock, invoices, calibration due.
2. **Approval tabs**
   Material requests, brand new parts, PRs.
3. **Waiting for your approval**
   Approve or reject in the row.
4. **Revenue**
   Invoice value by business unit.
5. **Purchases**
   PR / PO / delivered bars.
6. **Projects**
   Ongoing and upcoming.

> **RESULT**
> The manager's whole picture on one page.

ABOUT · DATA HEALTH

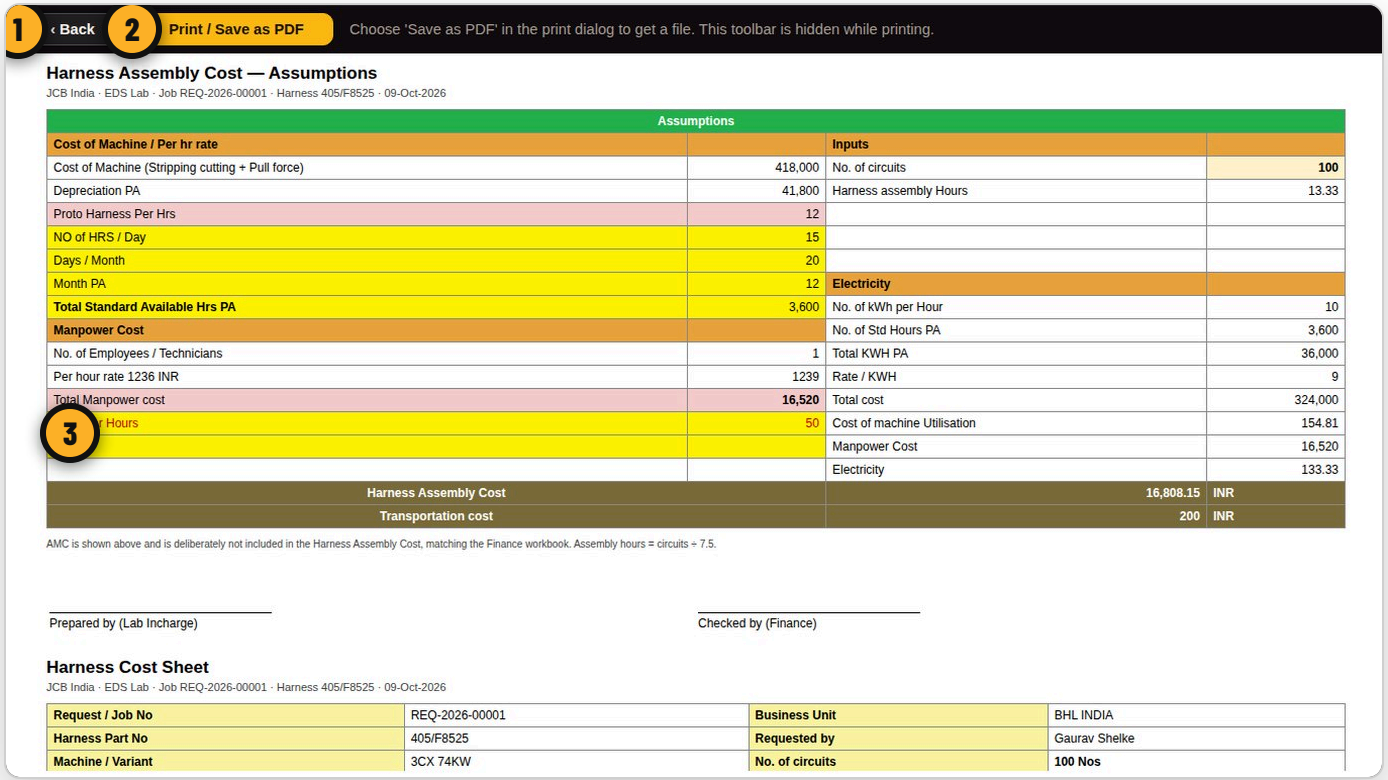


CLICK → WHAT HAPPENS

1. **How it works**
   Live stock, request to release, approvals, cost and invoices in four lines.
2. **Data Health**
   Rows in the sheet vs rows the app reads.
3. **Open System Check**
   The 18-row check.
4. **Re-read Now**
   Refresh every table.
5. **Check one part**
   Shows how one part number is read, to catch typing errors.
6. **Test Photo Storage**
   Confirms harness photos can be saved.

RESULT
Admins can prove the data is complete without opening the workbook.

PRINT / PDF



CLICK → WHAT HAPPENS

1. **Back**
   Returns to the invoice or cost sheet.
2. **Print / Save as PDF**
   Browser print: paper or PDF.
3. **Printable page**
   White page with the job card or cost sheet tables.

RESULT
Signed paper copy for the job file.